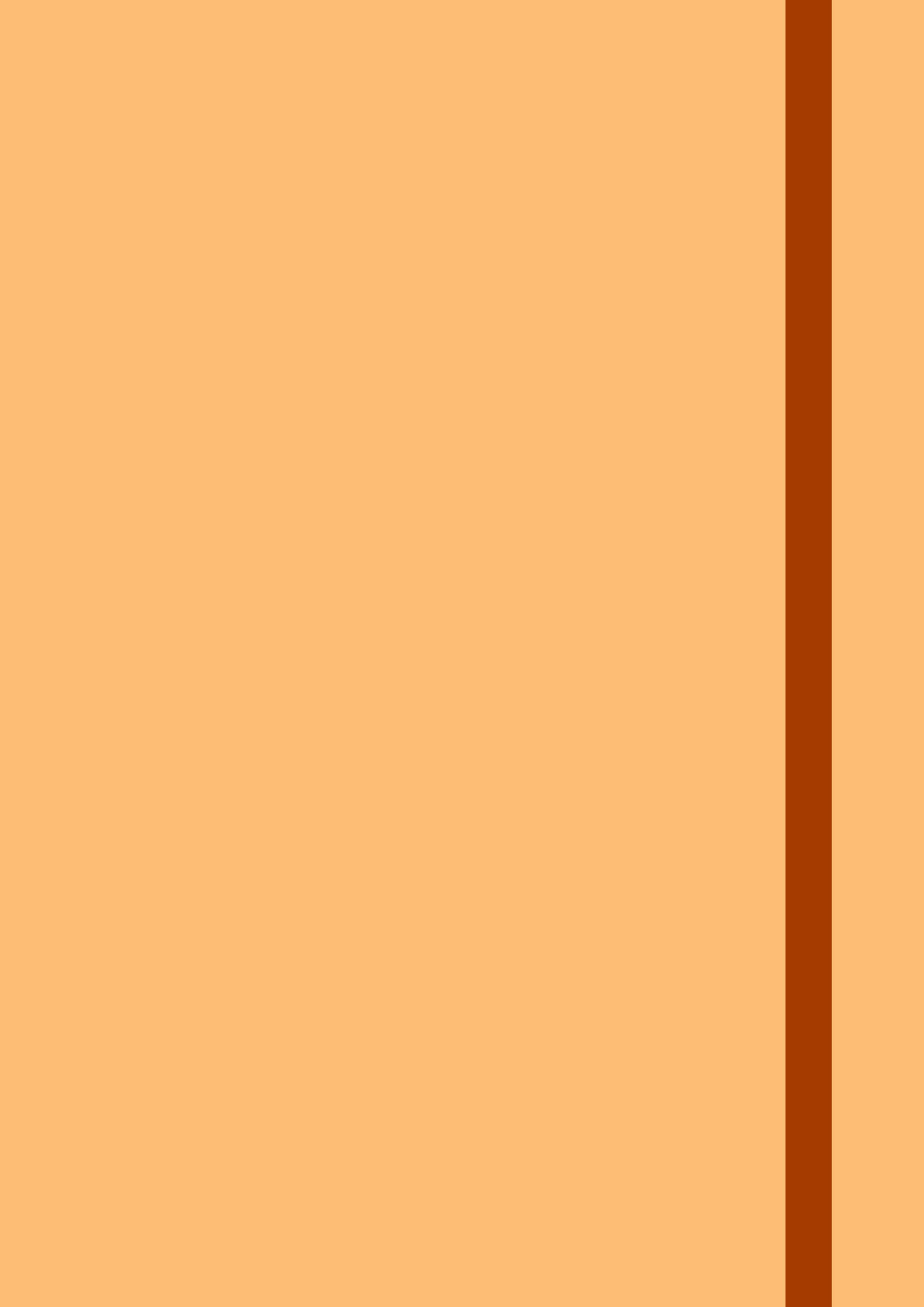




* pgm = portable graymap

P2 ~ format specifier

6 4 ~ resolution

255 ~ max pixel value

0
10
20
:
} pixel values

* ppm = portable pixmap

P3

6 4

255

0 0 0

10 10 10

20 20 20

:

red green blue
complement $\left\{ \begin{array}{ccc} R & G & B \\ C & M & Y \end{array} \right.$
cyan magenta yellow

0 255 255

= complement of 255 0 0

= complement of red

= cyan

* DIGITAL IMAGE = an array of numbers representing colours or intensities.

0 - 255

black white

GRAYSCALE: 1 2D channel 1 pixel = 8 bits

COLOUR: 3 2D channels 1 pixel = 24 bits

TYPES OF IMAGES

based on source $\left\{ \begin{array}{l} \text{radiation from EM spectrum} \\ \text{acoustic / ultrasonic / spectrogram} \\ \text{electronic} \\ \text{computer generated} \end{array} \right.$

based on light interaction $\left\{ \begin{array}{l} \text{reflection} \sim \text{info about object surface} \\ \text{emission} \sim \text{info about internal properties} \\ \text{absorption} \sim \text{info about internal structure} \end{array} \right.$

based on arrangement $\left\{ \begin{array}{l} \text{grayscale} \\ \text{rgb} \\ \text{multispectral} \\ \text{stereo} \\ \text{multi-view} \end{array} \right.$

DIGITAL IMAGE PROCESSING

= study and development of algorithms that operate on an image

to create new images / to retrieve its attributes

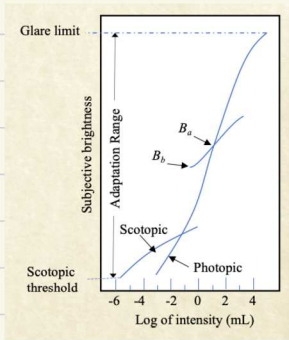
for consumption by human eyes / machine-based processes

IMAGE PROCESSING = pixel to pixel

COMPUTER VISION = pixel to symbol

COMPUTER GRAPHICS = symbol to pixel

BRIGHTNESS ADAPTATION

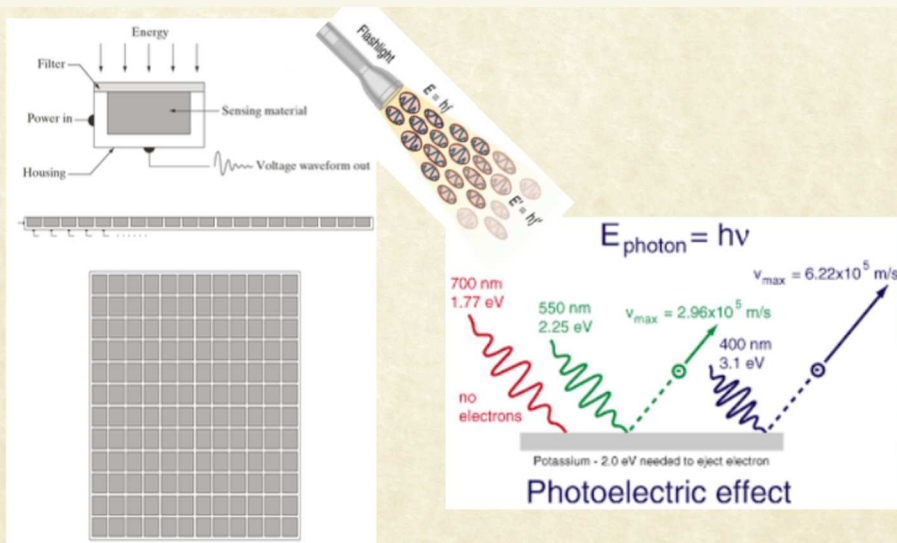


The human eye can adapt to a wide range of light intensities, but only a fraction at any given moment.

Adaptation shifts this range up or down depending on the ambient light level.

IMAGE SENSING AND ACQUISITION

- Single Sensor
- Line Sensor
- Array Sensor

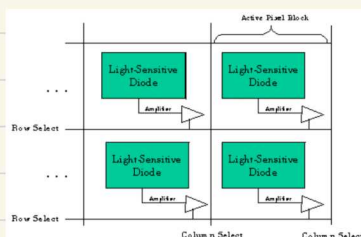


1. Light from scene → Filter → Sensing material
2. Photons hit the sensor and free electrons (photoelectric effect)
3. Electrons create a voltage waveform.
4. Read out these signals and convert them into digital values (pixel intensities)

Complementary Metal-Oxide-Semiconductor

CMOS Photoelectric Sensor

most common image sensor in phones and cameras today



1. Photons hit the pixel's photodiode
2. If photon energy > silicon bandgap, electrons are freed.
3. Freed electrons are stored in a tiny capacitor in the pixel during exposure.
4. Each pixel has its own transistor circuit to amplify and output the voltage.

CMOS is sensitive to LIGHT, not COLOUR.

Bayer Filter = 50% G, 25% R, 25% B ~ each pixel records only one colour component
 Demosaicing = Reconstruct the missing two colours (interpolated from neighboring pixels)

PIXEL SIZE MATTERS

~ even when the pixel count is the same, the image taken with the larger-sized pixels is clearer.

Larger pixels = more photons

= stronger signal / better signal-to-noise ratio

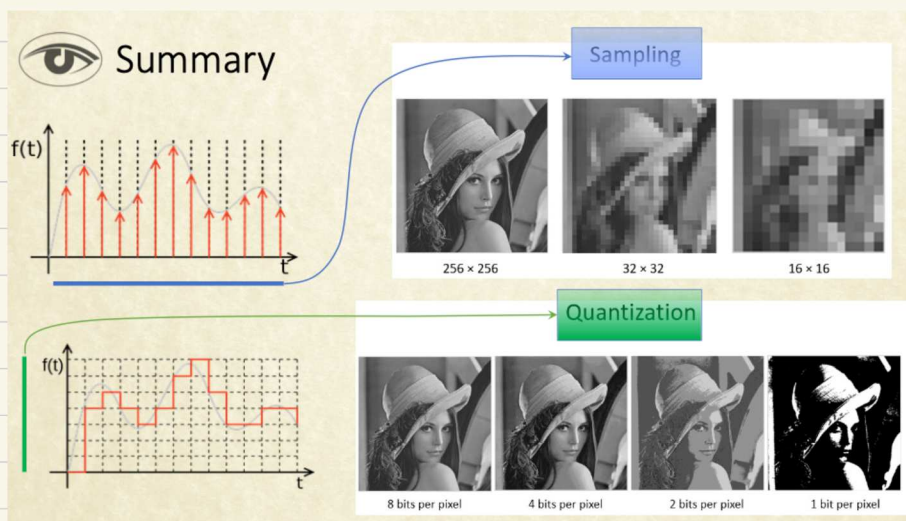
= better low-light performance

Analog signals are converted to digital signals via SAMPLING and QUANTISATION

spatial

intensity

Since we have limited # bits per pixel



SAMPLING

= the process of converting a continuous image $f(x,y)$ to a discrete representation $f_s[m,n]$ by recording the function values at points of intersection of a 2D grid.

$$f_s[m,n] = f(x,y) \cdot \sum_{m=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} \delta(x-mX_0, y-nY_0)$$
$$= \sum_{m=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} f(mX_0, nY_0) \delta(x-mX_0, y-nY_0)$$

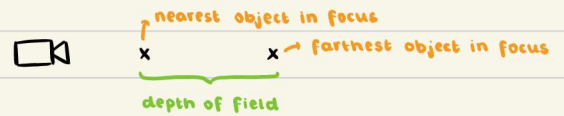
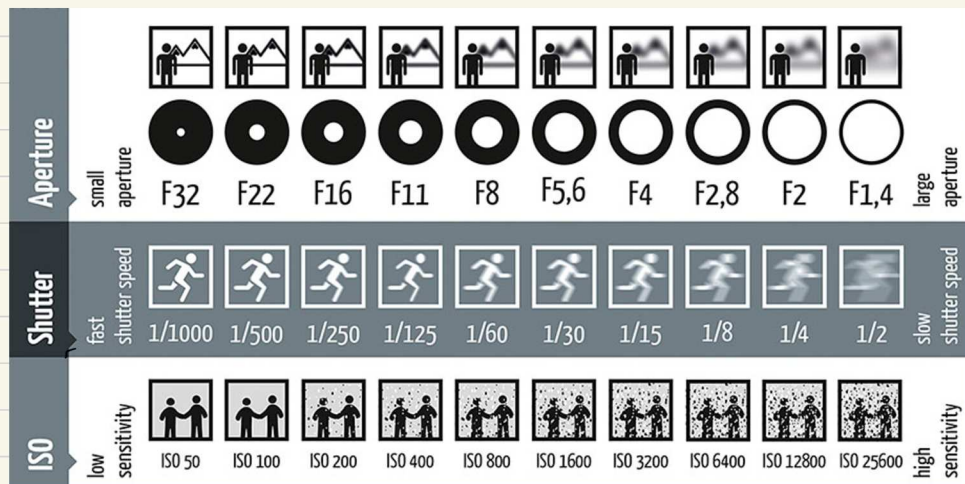
$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$

X_0, Y_0 = sampling distances / intervals

QUANTISATION

= in hardware, # of voltage levels = # of bits

TEMPORAL SAMPLING



APERTURE

= the size of the opening in the lens that lets light into the camera

large aperture = lets in a lot of light ; small aperture = lets in less light

EFFECT ON IMAGE → DEPTH OF FIELD

SHUTTER SPEED

= how long the camera shutter stays open to let light in

fast shutter = freezes motion , slow shutter = introduces motion blur

EFFECT ON IMAGE = motion blur + brightness

longer open = more light captured

ISO

= the sensitivity of the camera sensor to light

high ISO = very sensitive , useful in low light , more digital noise (grainy image)

low ISO = less sensitive , needs more daylight , clean smooth images

EFFECT ON IMAGE = brightness + noise/grain

CAMERAS

curtains

Two shutters for uniform illumination across the image from top to bottom.

open/close in opposite directions

First curtain opens to begin the exposure ; Second curtain closes to end the exposure

* Rolling Shutter Artifact (shearing effect)

In digital cameras, exposure sometimes doesn't happen for the entire image at once.

Instead, it is captured line by line.

Distortion occurs if the subject moves quickly or if the camera pans fast during this process.

~ straight lines may appear slanted.

~ fast-moving objects may look skewed.

- * Earlier film cameras used a rotating cylinder with a slit instead of flat curtains.
- * As the cylinder rolled, the slit allowed light to expose the film strip behind it.

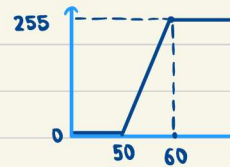
PIXEL PROCESSING / TRANSFORMATIONS

• Contrast Stretching / Normalisation

e.g. if min=50 and max=60

$$255 \times \left(\frac{I - \min}{\max - \min} \right) \quad \left. \vphantom{\frac{I - \min}{\max - \min}} \right\} \text{ b/w 0 and 255}$$

b/w 0 and 1

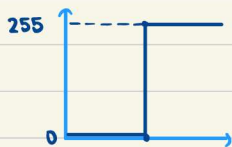


In general: $f_{ac}(a) = a_{\min} + (a - a_{\text{low}}) \times \frac{a_{\max} - a_{\min}}{a_{\text{high}} - a_{\text{low}}}$

original range: $a_{\text{low}} - a_{\text{high}}$

new range: $a_{\min} - a_{\max}$

• Binary Thresholding / Binarisation

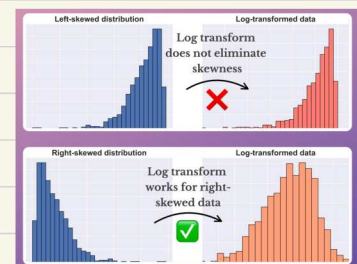
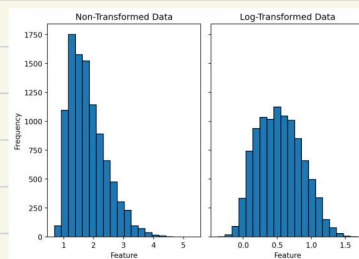
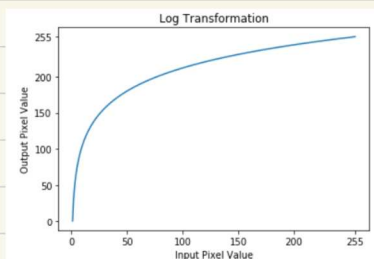


- * Most transformations are monotonically increasing functions.

• Log Transformation

$$s = T(r) = c \log(1+r)$$

to prevent log 0



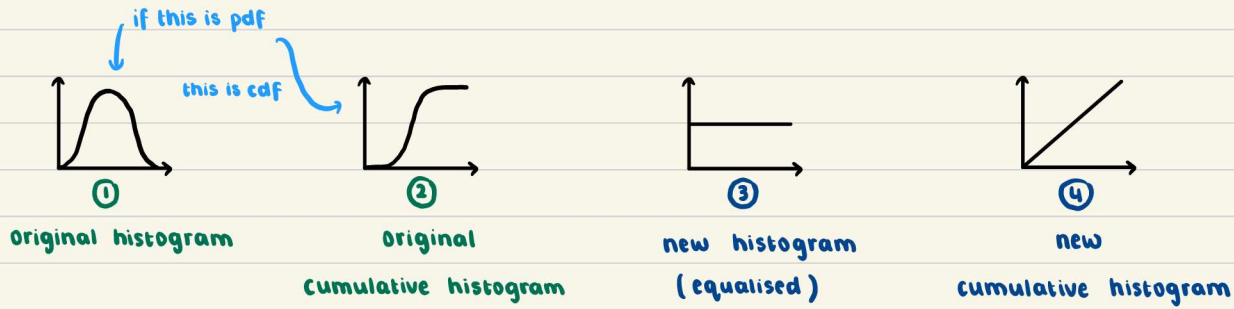
CONTRAST STRETCHING

- linearly map the pixel intensities from the current range to the new range.
- preserves the shape of the histogram, only expands / shifts it.

HISTOGRAM EQUALISATION

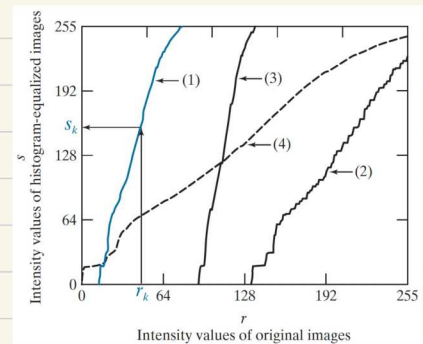
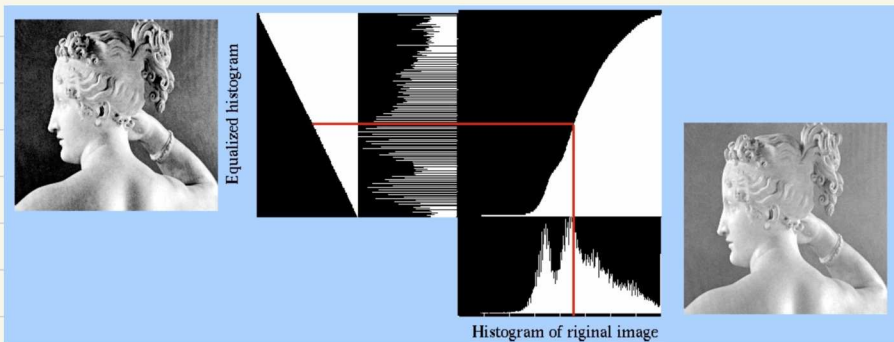
- redistribute intensities so that they spread UNIFORMLY across the entire range.

• Histogram Equalisation



MAIN IDEA: Apply ② to ① to get ③

Flatter (equalised) histogram $\Rightarrow$ More triangular cdf

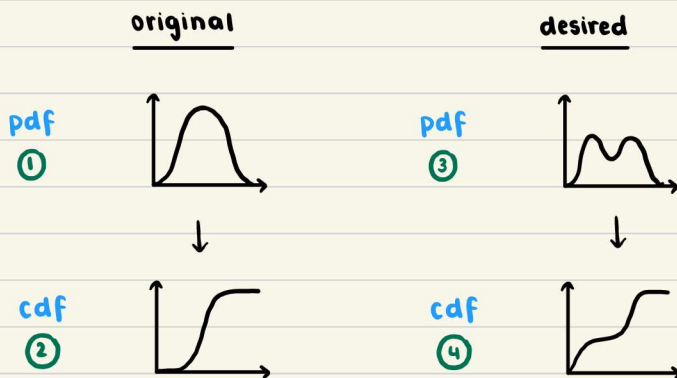


• Histogram Specification / Histogram Matching

$\rightarrow$ instead of just flattening the histogram (like equalisation), here we want the output image to have a desired histogram shape

$\rightarrow$ WE want to make the image look like a particular intensity distribution

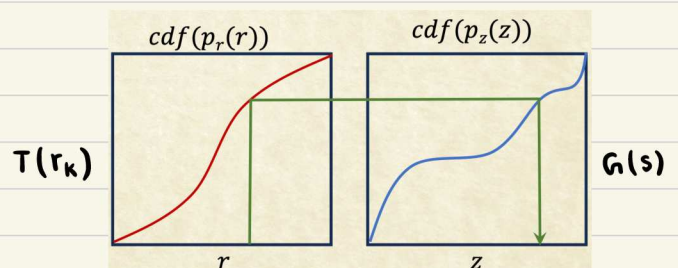
TO CHANGE FROM ONE HISTOGRAM TO ANOTHER:



1. Apply ② to ①
 $\rightarrow$ get flat histogram (equalisation)

2. Apply inverse of ④ to the flat histogram
 $\rightarrow$ get ③ (desired histogram)

1. Obtain $p_r(r)$ from input image.
2. Compute cdf $S = T(r_k)$ from $p_r(r)$.
3. Obtain $p_z(z)$ from target image.
4. Compute cdf $z = G^{-1}(s)$ from $p_z(z)$.
5. Combine $T(r_k)$ and $G^{-1}(s)$ to obtain the mapping
 $z = G^{-1}(T(r_k))$



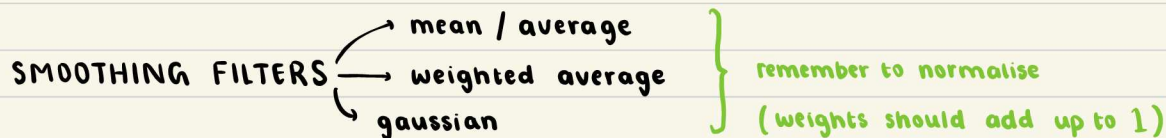
- * In raw (unnormalised) form, histograms are image-size dependent
- * When normalised, they do not depend on image-size
 - useful statistical representation of image intensities

GLOBAL HISTOGRAM EQUALISATION

- ~ uses the histogram of the entire image to compute the transformation function.
- ~ works well if the entire image has poor contrast
 - if contrast varies across regions → may not enhance all areas equally.
 - can cause over-enhancement in uniform regions (noise amplification)

LOCAL HISTOGRAM EQUALISATION

- ~ instead of one global histogram, the image is divided into small regions
- ~ enhances local contrast, helps reveal hidden details
 - preserves details across regions of different brightness



Higher smoothing effect → larger kernel size / more iterations

$\frac{1}{9} \times$	1	1	1
	1	1	1
	1	1	1
Standard average			

$\frac{1}{16} \times$	1	2	1
	2	4	2
	1	2	1
Weighted Average			

GAUSSIAN SMOOTHING

Specify either sigma or window size

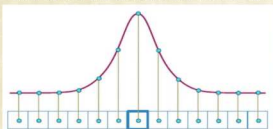
take window size
until values reach 0

take whatever
sigma fits

* COMPUTING GAUSSIAN FILTER COEFFICIENTS

Binomial coefficients approximate the Gaussian distribution ⇒ use Pascal's triangle

Index N	Coefficients	Sum=2 ^N
0	1	1
1	1 1	2
2	1 2 1	4
3	1 3 3 1	8
4	1 4 6 4 1	16
5	1 5 10 10 5 1	32
6	1 6 15 20 15 6 1	64
7	1 7 21 35 35 21 7 1	128
8	1 8 28 56 70 56 28 8 1	256
9	1 9 36 84 126 126 84 36 9 1	512
10	1 10 45 120 210 252 210 120 45 10 1	1024



1. Pick a row from Pascal's triangle based on filter size.
2. Normalise to get 1D gaussian kernel.
3. Take outer product with itself to get 2D gaussian kernel.

$$(u \otimes v)_{ij} = u_i v_j$$
4. Normalise again to get final gaussian filter.

e.g., $s = 7 \times 7$

$\frac{1}{64} \times$	1	6	15	20	15	6	1
-----------------------	---	---	----	----	----	---	---

e.g., $s = 7 \times 7$

$\frac{1}{4096} \times$	1	6	15	20	15	6	1
	6	36	90	120	90	36	6
	15	90	225	300	225	90	15
	20	120	300	400	300	120	20
	15	90	225	300	225	90	15
	6	36	90	120	90	36	6
	1	6	15	20	15	6	1

EDGE DETECTION

EDGE $\rightarrow$ intensity changes rapidly along one direction

$\rightarrow$ derivative in one direction should be high + derivatives in other directions should be low.

$$* f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$* \frac{\partial f(x,y)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

$$* \frac{\partial f(x,y)}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

Digital images are sampled and quantised $\Rightarrow$ discrete.

So the smallest Δx possible = 1

$\therefore$ In digital images, derivative = difference

$$\Rightarrow \frac{\partial f(x,y)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x} = f[x+1, y] - f[x, y]$$

$\hookrightarrow$ pixel values $\checkmark$

• 1st derivative = difference between two adjacent pixel values.

• 2nd derivative = difference between two adjacent 1st derivative values.

$$= (f[x+1, y] - f[x, y]) - (f[x, y] - f[x-1, y])$$

6	6	6	5	4	3	2	1	1	1	1	1	6	6	6	6	$\rightarrow x$
0	0	0	-1	-1	-1	-1	0	0	0	0	0	5	0	0	0	$\rightarrow x$
0	0	-1	0	0	0	0	1	0	0	0	0	5	-5	0	0	$\rightarrow x$

• 1st derivative as symmetric difference:

$$\frac{\partial f}{\partial x} = \frac{f(x+\Delta x, y) - f(x-\Delta x, y)}{2\Delta x}$$

$$\equiv \frac{1}{2} \begin{bmatrix} -1 & & +1 \end{bmatrix}$$

$$\frac{\partial f}{\partial y} = \frac{f(x, y+\Delta y) - f(x, y-\Delta y)}{2\Delta y}$$

$$\equiv \frac{1}{2} \begin{bmatrix} +1 \\ \\ -1 \end{bmatrix}$$

Gradient gives the direction of steepest ascent (direction of most rapid increase in intensity)

$$\nabla f = (f_x \ f_y) = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right) \sim \text{a vector}$$

Gradient image will have TWO numbers corresponding to each pixel.

EDGE IS PERPENDICULAR TO GRADIENT.

Prewitt Edge Filter $\rightarrow$ takes care of noise

$\rightarrow$ takes symmetric difference

$\rightarrow$ lightweight (just 0s and 1s)

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

Sobel filter: $\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} -2 & 0 & 2 \end{bmatrix} \rightarrow$ more weight to the center row

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

• LAPLACIAN FILTER

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

→ peak when dark to bright

→ trough when bright to dark

UNSHARP MASKING

$$I' = I + \underset{\text{laplacian}}{\nabla^2 I}$$

HIGH BOOST FILTERING

original - smoothened = detail

original + detail = sharpened

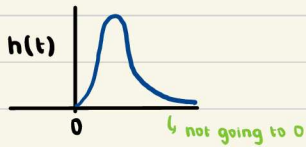
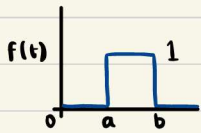
LINEAR SHIFT INVARIANT SYSTEMS

linear combination of inputs results in the same linear combination of outputs

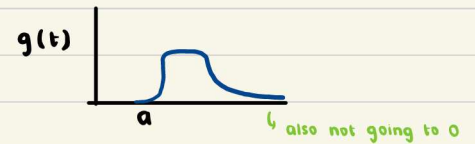
shift invariant $\times$ shift equivariant $\checkmark$

LSIS $\equiv$ convolutions

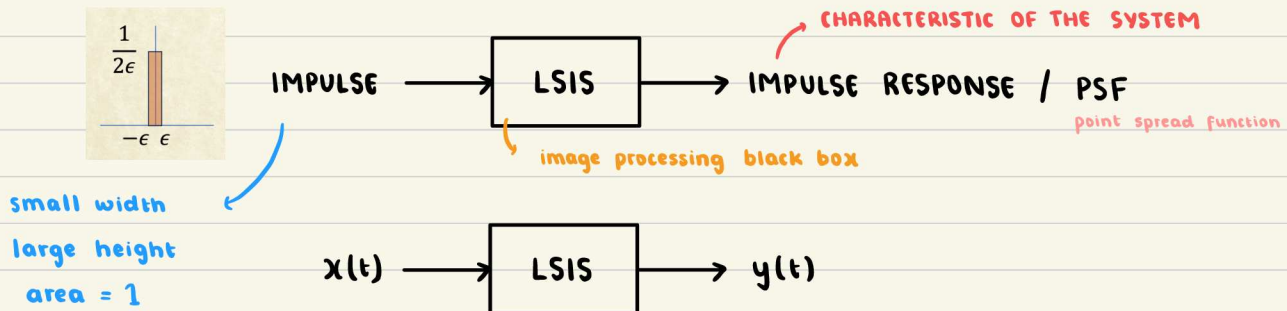
mathematically, convolution : $f(t) * h(t) = \int_{-\infty}^{\infty} f(z) h(t-z) dz = g(t)$



$h(t-z) = h(-z+t)$
flip and shift by t



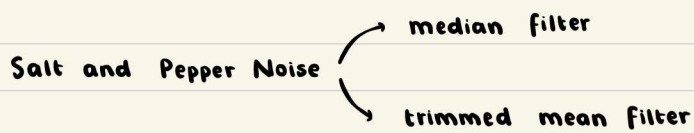
$\int f(z+s) \cdot h(t-z+s) dz \Rightarrow$ Put $y = t+z \rightarrow dy = dz$ (SHIFT INVARIANT)



If we convolve $x(t)$ with the impulse response of the system, we get the output $y(t)$.

$\Rightarrow$ We can reveal what is happening inside by just sending one impulse.





• TRIMMED MEAN FILTER

Say the filter is 3×3 .

Sort all 9 pixel values.

Trim 3 values on either side.

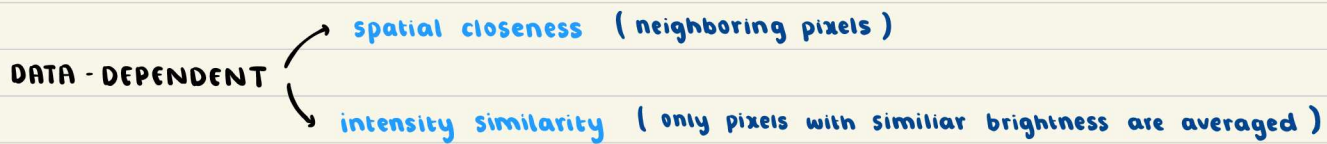
Take the average of the remaining 3.

* Min, Max, Median $\rightarrow$ non-linear filters, cannot model as weights

BILATERAL FILTER

$\rightarrow$ smooths images while preserving edges

$\rightarrow$ kernel depends on image content



We define the filtered output $g(i,j)$ at pixel (i,j) as : $g(i,j) = \frac{\sum_{k,l} f(k,l) \cdot w(i,j,k,l)}{\sum_{k,l} w(i,j,k,l)}$

$f(k,l)$: intensity of neighbour pixel at (k,l)

$w(i,j,k,l)$: weight given to neighbor at (k,l)

The weighting coefficient $w(i,j,k,l)$ is a product of two kernels:

DOMAIN KERNEL
(spatial closeness)

$$d(i,j,k,l) = \exp\left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_d^2}\right)$$

RANGE KERNEL
(intensity similarity)

$$r(i,j,k,l) = \exp\left(-\frac{\|f(i,j) - f(k,l)\|^2}{2\sigma_r^2}\right)$$

$$w(i,j,k,l) = \exp\left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_d^2} - \frac{\|f(i,j) - f(k,l)\|^2}{2\sigma_r^2}\right)$$

EDGE DETECTION UNDER NOISE

→ smooth with a Gaussian kernel first, then take derivative $\Rightarrow \frac{d}{dx} (f * g)$

Derivative Theorem of Convolution

In image processing, differentiation = convolving with a kernel that approximates the derivative (Prewitt, Sobel, etc.)

"Differentiation is convolution and convolution is associative" $\Rightarrow \frac{d}{dx} (f * g) = f * \left(\frac{d}{dx} g \right)$
($f * g$) * h = f * (g * h) Convolution is commutative too

⇒ Differentiating after smoothing is the same as smoothing with the derivative of the kernel.

∴ Derivative of Gaussian is good for edge detection under noise.

LAPLACIAN OF GAUSSIAN

→ combines Gaussian smoothing with the Laplacian operator to detect edges and other features in an image while reducing sensitivity to noise.

→ can be approximated by the difference of two Gaussians with different σ values.

$$LoG \approx DoG$$

IMPULSE

- very small time duration, very large height
- total area under the curve is 1

Any signal can be thought of as a sum of scaled and shifted impulse functions.

When it is passed through a linear shift invariant system,

the result will be the sum of scaled and shifted impulse responses.

RANK OF A SEPARABLE KERNEL = 1

A separable kernel means it can be written as the outer product of two vectors $K = uv^T$

Both of these row/column vectors has rank 1 (assuming non-zero)

Since $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$

$\Rightarrow \text{rank}(K) = \text{rank}(uv^T) \leq \min(1, 1) = 1$

PERIODIC SIGNALS

time domain

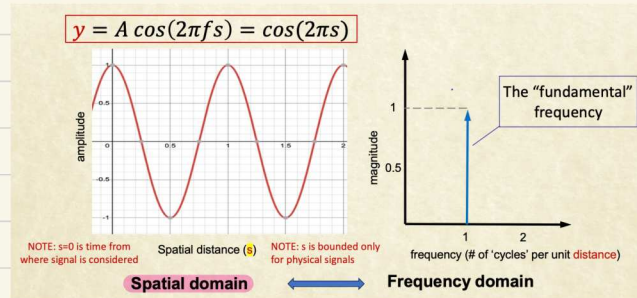
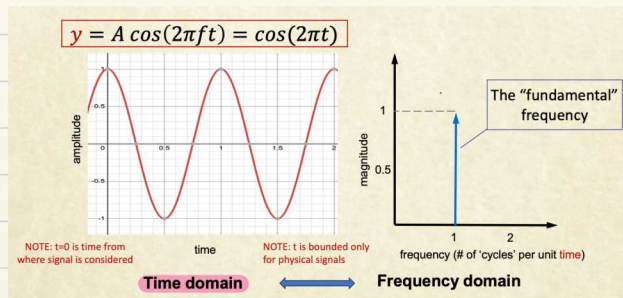
spatial domain

frequency = cycles/sec or cycles / unit length

angular frequency = radians / sec or radians / unit length

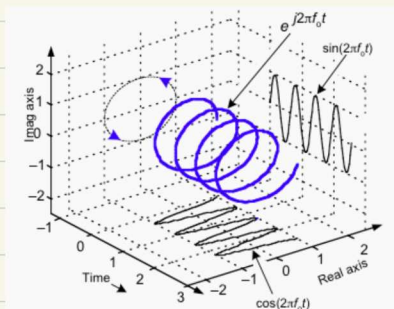
period = seconds / cycle or distance / cycle

$$x(t) = A \cos(\omega t) = A \cos(2\pi f t) = A \cos\left(\frac{2\pi t}{T}\right)$$



Frequency-domain view shows which frequencies exist in the signal.

Fundamental frequency = lowest frequency at which the signal repeats itself.



$$e^{it} = \cos t + i \sin t$$

Along the time axis, the signal progresses.

Along the real axis, you see the cosine part.

Along the imaginary axis, you see the sine part.

IMPULSES AND THEIR SIFTING PROPERTIES

$$\text{unit impulse} = \delta(t) = \begin{cases} \infty & \text{if } t=0 \\ 0 & \text{if } t \neq 0 \end{cases}, \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\text{Sifting Property : } \int_{-\infty}^{\infty} f(t) \cdot \delta(t) dt = f(0), \quad \int_{-\infty}^{\infty} f(t) \cdot \delta(t-t_0) dt = f(t_0)$$

$$\text{unit discrete impulse, } \delta(x) = \begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } x \neq 0 \end{cases}, \quad \sum_{x=-\infty}^{\infty} \delta(x) = 1$$

$$\text{Sifting Property : } \sum_{x=-\infty}^{\infty} f(x) \cdot \delta(x) = f(0), \quad \sum_{x=-\infty}^{\infty} f(x) \cdot \delta(x-x_0) = f(x_0)$$

PROOF OF THE SIFTING PROPERTY

$$\int_{-\infty}^{\infty} f(t) \cdot \delta(t-t_0) dt = \int_{t_0-\epsilon}^{t_0+\epsilon} f(t) \cdot \delta(t-t_0) dt \approx f(t_0) \cdot \int_{-\infty}^{\infty} \delta(t-t_0) dt = f(t_0) \cdot 1 = f(t_0)$$

FOURIER SERIES

~ for periodic functions

$$f(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{j \frac{2\pi n t}{T}}, \quad c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot e^{-j \frac{2\pi n t}{T}} dt$$

FOURIER TRANSFORM

~ for non-periodic functions

$$F\{f(t)\} = F(\mu) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j2\pi \mu t} dt \quad \text{frequency} = \text{forward Fourier transform}$$

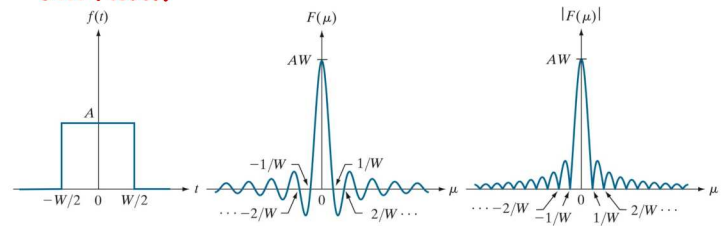
$$F^{-1}\{F(\mu)\} = f(t) = \int_{-\infty}^{\infty} F(\mu) \cdot e^{j2\pi \mu t} d\mu = \text{inverse Fourier transform}$$

NOTE: $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$, $\text{sinc}(m) = \frac{\sin(\pi m)}{(\pi m)}$

* Fourier Transform of a simple continuous function:

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi \mu t} dt = \int_{-W/2}^{W/2} A e^{-j2\pi \mu t} dt \\ &= \frac{-A}{j2\pi \mu} \left[e^{-j2\pi \mu t} \right]_{-W/2}^{W/2} = \frac{-A}{j2\pi \mu} \left[e^{-j\pi \mu W} - e^{j\pi \mu W} \right] \\ &= \frac{A}{j2\pi \mu} \left[e^{j\pi \mu W} - e^{-j\pi \mu W} \right] \\ &= AW \frac{\sin(\pi \mu W)}{(\pi \mu W)} \end{aligned}$$

BOX FILTER



a b c

FIGURE 4.4 (a) A box function, (b) its Fourier transform, and (c) its spectrum. All functions extend to infinity in both directions. Note the inverse relationship between the width, W , of the function and the zeros of the transform.

* Fourier Transform of an impulse:

$$F\{\delta(t)\} = F(\mu) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j2\pi \mu t} dt = e^{-j2\pi \mu} \quad (\text{constant})$$

$$F\{\delta(t-t_0)\} = F(\mu) = \int_{-\infty}^{\infty} \delta(t-t_0) \cdot e^{-j2\pi \mu t} dt = e^{-j2\pi \mu t_0}$$

using sifting property

Fourier Transform of an impulse train :

- If a function $f(t)$ has the Fourier transform $F(\mathcal{M})$, then evaluating this function at t , $F(t)$, must have the transform $f(-\mathcal{M})$.

Fourier transform of $\delta(t-t_0)$ is $e^{-j2\pi\mathcal{M}t_0}$

$\Rightarrow$ Fourier transform of $e^{-j2\pi\mathcal{M}t_0}$ is $\delta(-\mathcal{M}-t_0)$

Put $-t_0 = a$

Fourier transform of $e^{j2\pi a t}$ is $\delta(-\mathcal{M}+a) = \delta(\mathcal{M}-a)$

$$f(t) = \delta(t-t_0)$$

$$F(\mathcal{M}) = e^{-j2\pi\mathcal{M}t_0} \Rightarrow F(t) = e^{-j2\pi t t_0}$$

$$f(-\mathcal{M}) = \delta(-\mathcal{M}-t_0)$$

Put $t_0 = -a$

$$F(t) = e^{j2\pi a t} \Rightarrow f(-\mathcal{M}) = \delta(-\mathcal{M}+a) = \delta(\mathcal{M}-a)$$

IMPULSE TRAIN = the sum of infinitely many impulses ΔT units apart

$$S_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} \delta(t-k\Delta T)$$

Since the impulse train is periodic with period ΔT , it can be expressed as a Fourier series:

$$S_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{j\frac{2\pi n}{\Delta T} t} \quad , \quad c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} S_{\Delta T}(t) \cdot e^{-j\frac{2\pi n}{\Delta T} t} dt$$

The integral in the interval $[-\Delta T/2, \Delta T/2]$ encompasses only the impulse located at the origin.

$$c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) \cdot e^{-j\frac{2\pi n}{\Delta T} t} dt = \frac{1}{\Delta T} e^0 = \frac{1}{\Delta T}$$

$$S_{\Delta T}(t) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T} t}$$

$$F\left\{e^{j\frac{2\pi n}{\Delta T} t}\right\} = \delta(\mathcal{M} - \frac{n}{\Delta T}) \Rightarrow F\{S_{\Delta T}(t)\} = F\left\{\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T} t}\right\} = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta(\mathcal{M} - \frac{n}{\Delta T})$$

$\therefore$ The Fourier transform of an impulse train with period ΔT is also an impulse train with period $\frac{1}{\Delta T}$

CONVOLUTION THEOREM

$$(f * h)(t) = \int_{-\infty}^{\infty} f(\tau) \cdot \underbrace{h(t-\tau)}_{\text{flipping + displacement}} d\tau$$

$$(f * h)(t) \iff (F \cdot H)(\mathcal{M})$$

convolution

multiplication

spatial domain

frequency domain

AND

$$(f \cdot h)(t) \iff (F * H)(\mathcal{M})$$

multiplication

convolution

spatial domain

frequency domain

The Laplace transform of a convolution is the product of the Laplace transforms of the individual functions.

PROOF:

$$\begin{aligned}
 F\{(f * g)(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(z) \cdot h(t-z) dz \right] e^{-j2\pi\mu t} dt \\
 &= \int_{-\infty}^{\infty} f(z) \left[\int_{-\infty}^{\infty} h(t-z) e^{-j2\pi\mu t} dt \right] dz \\
 &\quad \text{Fourier transform of } h(t-z) \\
 &\quad = H(\mu) \cdot e^{-j2\pi\mu z} \longrightarrow \\
 &= H(\mu) \cdot \int_{-\infty}^{\infty} f(z) \cdot e^{-j2\pi\mu z} dz = H(\mu) \cdot F(\mu)
 \end{aligned}$$

* PROOF

$$F\{h(t-z)\} = \int_{-\infty}^{\infty} h(t-z) \cdot e^{-j2\pi\mu t} dt$$

$$(\text{Put } u=t-z \Rightarrow du=dt)$$

$$= \int_{-\infty}^{\infty} h(u) \cdot e^{-j2\pi\mu(u+z)} du = e^{-j2\pi\mu z} \cdot \int_{-\infty}^{\infty} h(u) \cdot e^{-j2\pi\mu u} du$$

$$= H(\mu) \cdot e^{-j2\pi\mu z}$$

SAMPLING

= multiplying by a sampling function equal to a train of impulses ΔT units apart.

$$\tilde{f}(t) = f(t) \cdot S_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t) \cdot \delta(t - n\Delta T)$$

* Fourier transform of the sampled function:

$$\tilde{F}(\mu) = F\{\tilde{f}(t)\} = F\{f(t) \cdot S_{\Delta T}(t)\} = (F * S)(\mu), \quad S(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta(\mu - \frac{n}{\Delta T})$$

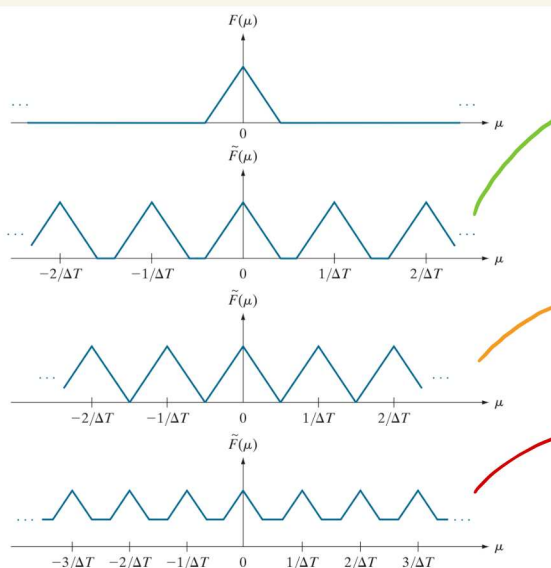
$$\tilde{F}(\mu) = (F * S)(\mu) = \int_{-\infty}^{\infty} F(z) \cdot S(\mu - z) dz = \frac{1}{\Delta T} \int_{-\infty}^{\infty} F(z) \cdot \sum_{n=-\infty}^{\infty} \delta(\mu - z - \frac{n}{\Delta T}) dz = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} F(z) \cdot \delta(\mu - z - \frac{n}{\Delta T}) dz$$

$$= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F(\mu - \frac{n}{\Delta T})$$

↑ sampling rate

The Fourier transform $\tilde{F}(\mu)$ of the sampled function $\tilde{f}(t)$

= an infinite, periodic sequence of copies of the transform of the original, continuous function.



OVER-SAMPLING

= sampling rate is high enough to provide sufficient separation b/w periods and preserve the integrity (perfect copies) of $F(\mu)$

CRITICALLY SAMPLING

= sampling rate is just enough to preserve $F(\mu)$

UNDER-SAMPLING

= sampling rate is below the minimum required to maintain distinct copies of $F(\mu)$

= fails to preserve the original transform

SAMPLING THEOREM

BAND-LIMITED FUNCTION

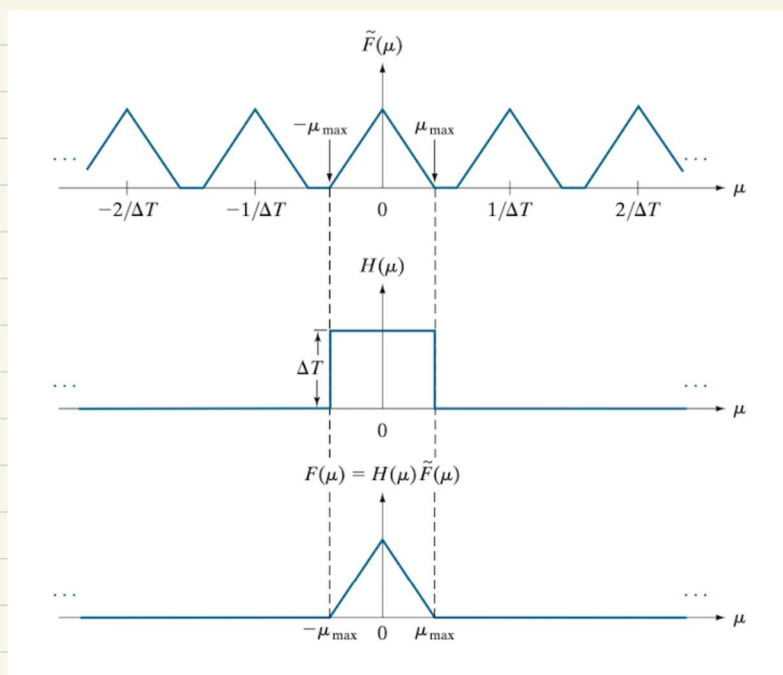
= a function $f(t)$ whose Fourier transform is zero for values of frequencies outside a finite interval (band) $[-\mathcal{M}_{\max}, \mathcal{M}_{\max}]$ about the origin.

We can recover $f(t)$ from its samples if we can isolate a single copy of $F(\mu)$ from the periodic sequence of copies of this function contained in $\tilde{F}(\mu)$, the transform of the sampled function $\tilde{f}(t)$.

We can recover $f(t)$ from that single period by taking its inverse Fourier transform.

A continuous band-limited function can be recovered completely from a set of its samples if the samples are acquired at a rate exceeding twice the highest frequency content of the function. } **SAMPLING THEOREM**
sampling rate = $\frac{1}{\Delta T} > 2\mathcal{M}_{\max}$

A sampling rate exactly equal to twice the highest frequency is called the Nyquist rate.



IDEAL LOWPASS FILTER

$$H(\mu) = \begin{cases} \Delta T & -\mathcal{M}_{\max} \leq \mu \leq \mathcal{M}_{\max} \\ 0 & \text{otherwise} \end{cases}$$

isolates the period centered on the origin

$$F(\mu) = H(\mu) \tilde{F}(\mu)$$

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

ALIASING

= sampling phenomena that cause different signals to become indistinguishable from one another after sampling.
aliased pair

REASON: We used a sampling rate that was too coarse
→ The functions were under-sampled.

Sampling rate < Nyquist rate

= periods of the Fourier transform overlap

= impossible to isolate a single period of the transform

= inverse transform yields a function different from the original

* Aliasing is always present in sampled signals

- even if the original sampled function is band-limited, infinite frequency components are introduced the moment we limit the duration of the function.

$$h(t) = \begin{cases} 1 & 0 \leq t \leq T \\ 0 & \text{otherwise} \end{cases} \quad \begin{aligned} \mathcal{F}\{h(t)\} &= H(\omega) = \text{has infinite frequency components in both directions} \\ \therefore \text{since } f(t) \cdot h(t) &= (F * H)(\omega) \\ \Rightarrow \text{once duration is limited} &\rightarrow \text{no longer band-limited} \end{aligned}$$

ANTI-ALIASING

= lowpass filtering the input function to attenuate the higher frequencies BEFORE SMOOTHING

FUNCTION RECONSTRUCTION FROM SAMPLED DATA

Reconstructing a function from a set of its samples

= Interpolating between the samples

$$F(\omega) = H(\omega) \cdot \tilde{F}(\omega)$$

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \mathcal{F}^{-1}\{H(\omega) \cdot \tilde{F}(\omega)\} = (h * \tilde{f})(t) = \int_{-\infty}^{\infty} h(\tau) \tilde{f}(t - \tau) d\tau$$

$$\tilde{f}(t) = f(t) \cdot S_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(n\Delta T) \cdot \delta(t - n\Delta T) \Rightarrow \tilde{f}(t - \tau) = \sum_{n=-\infty}^{\infty} f(n\Delta T) \cdot \delta(t - \tau - n\Delta T)$$

$$f(t) = \int_{-\infty}^{\infty} h(\tau) \left[\sum_{n=-\infty}^{\infty} f(n\Delta T) \cdot \delta(t - \tau - n\Delta T) \right] d\tau = \sum_{n=-\infty}^{\infty} f(n\Delta T) \underbrace{\int_{-\infty}^{\infty} h(\tau) \delta(t - \tau - n\Delta T) d\tau}_{\text{apply sifting property}} = \sum_{n=-\infty}^{\infty} f(n\Delta T) h(t - n\Delta T)$$

$$\begin{aligned} h(t) &= \int_{-\omega_{\max}}^{\omega_{\max}} \Delta T \cdot e^{j2\pi\omega t} d\omega = \Delta T \int_{-1/2\Delta T}^{1/2\Delta T} e^{j2\pi\omega t} d\omega = \frac{\Delta T}{j2\pi t} \left[e^{j2\pi\omega t} \right]_{-1/2\Delta T}^{1/2\Delta T} = \frac{\Delta T}{j2\pi t} \left[e^{j\pi t/\Delta T} - e^{-j\pi t/\Delta T} \right] \\ &= \frac{\Delta T}{\pi t} \sin(\pi t/\Delta T) = \text{sinc}(t/\Delta T) \Rightarrow h(t - n\Delta T) = \text{sinc}\left(\frac{t - n\Delta T}{\Delta T}\right) \end{aligned}$$

$$f(t) = \sum_{n=-\infty}^{\infty} f(n\Delta T) \text{sinc}\left(\frac{t - n\Delta T}{\Delta T}\right)$$

- The reconstructed function is identically equal to the sample values at multiple integer increments of ΔT .
- Between sample points, values of $f(t)$ are interpolations formed by the sum of sinc functions.

DISCRETE FOURIER TRANSFORM

We have the transform $\tilde{F}(\omega)$ of sampled data in terms of the transform of the original function.

We want an expression for $\tilde{F}(\omega)$ in terms of the sampled function $\tilde{f}(t)$ itself.

$$\tilde{F}(\omega) = \int_{-\infty}^{\infty} \tilde{f}(t) \cdot e^{-j2\pi\omega t} dt = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(t) \cdot \delta(t-n\Delta T) e^{-j2\pi\omega t} dt = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cdot \delta(t-n\Delta T) e^{-j2\pi\omega t} dt = \sum_{n=-\infty}^{\infty} f_n \cdot e^{-j2\pi\omega n\Delta T}$$

$$f_n = f(n\Delta T)$$

$\tilde{F}(\omega)$ is continuous and infinitely periodic with period $1/\Delta T$.

All we need to characterise $\tilde{F}(\omega)$ is one period.

Sampling one period of this function is the basis of the DFT.

Suppose we want to obtain M equally spaced samples of $\tilde{F}(\omega)$ taken over the one period interval from $\omega=0$ to $\omega=1/\Delta T$.

We take samples at the following frequencies:

$$\omega = \frac{m}{M\Delta T} \quad m = 0, 1, 2, \dots, M-1$$

$$\tilde{F}(\omega) = \sum_{n=-\infty}^{\infty} f_n \cdot e^{-j2\pi\omega n\Delta T} = \tilde{F}\left(\frac{m}{M\Delta T}\right) = \sum_{n=-\infty}^{\infty} f_n \cdot e^{-j2\pi \frac{m}{M\Delta T} n\Delta T} = \sum_{n=0}^{M-1} f_n \cdot e^{-j2\pi mn/M}$$

we don't need infinite samples

$$F_m = \sum_{n=0}^{M-1} f_n \cdot e^{-j2\pi mn/M} \Rightarrow \text{forward Discrete Fourier Transform}$$

Given $\{F_m\}$, we can recover the sample set $\{f_m\}$

$$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m \cdot e^{j2\pi mn/M} \Rightarrow \text{inverse Discrete Fourier Transform}$$

It is more intuitive, especially in two dimensions, to use the notation x and y for image coordinate variables and u and v for frequency variables.

$$\left. \begin{aligned} F(u) &= \sum_{x=0}^{M-1} f(x) \cdot e^{-j2\pi ux/M} & u &= 0, 1, 2, \dots, M-1 & F(u) &\equiv F_m \\ f(x) &= \frac{1}{M} \sum_{u=0}^{M-1} F(u) \cdot e^{j2\pi ux/M} & x &= 0, 1, 2, \dots, M-1 & F(x) &\equiv F_n \end{aligned} \right\} \text{1D DFT pair}$$

* Both the forward and inverse discrete transforms are infinitely periodic, with period M .

$$F(u) = F(u+KM) \quad f(x) = f(x+KM)$$

• Discrete equivalent of the 1D convolution:

$$f(x) * h(x) = \sum_{m=0}^{M-1} f(m) \cdot h(x-m) \quad x = 0, 1, 2, \dots, M-1$$

EXAMPLE 4.4: The mechanics of computing the DFT.

Figure 4.12(a) shows four samples of a continuous function, $f(t)$, taken ΔT units apart. Figure 4.12(b) shows the samples in the x -domain. The values of x are 0, 1, 2, and 3, which refer to the number of the samples in sequence, counting up from 0. For example, $f(2) = f(t_0 + 2\Delta T)$, the third sample of $f(t)$.

From Eq. (4-44), the first value of $F(u)$ [i.e., $F(0)$] is

$$F(0) = \sum_{x=0}^3 f(x) = [f(0) + f(1) + f(2) + f(3)] = 1 + 2 + 4 + 4 = 11$$

The next value of $F(u)$ is

$$F(1) = \sum_{x=0}^3 f(x) e^{-j2\pi(1)x/4} = 1e^0 + 2e^{-j\pi/2} + 4e^{-j\pi} + 4e^{-j3\pi/2} = -3 + 2j$$

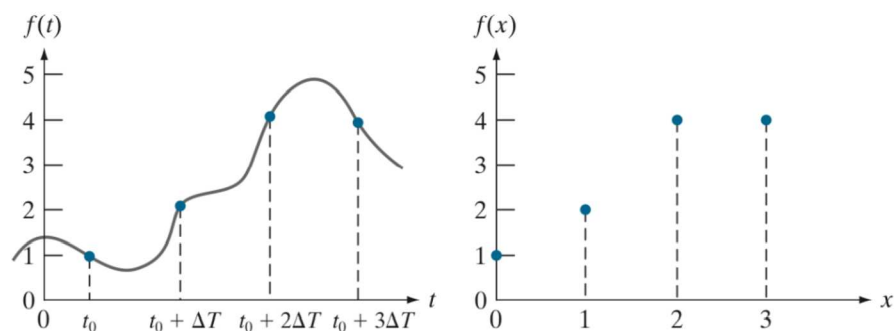
Similarly, $F(2) = -(1 + 0j)$ and $F(3) = -(3 + 2j)$. **Observe that all values of $f(x)$ are used in computing each value of $F(u)$.**

If we were given $F(u)$ instead, and were asked to compute its inverse, we would proceed in the same manner, but using the inverse Fourier transform. For instance,

$$f(0) = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j2\pi u(0)} = \frac{1}{4} \sum_{u=0}^3 F(u) = \frac{1}{4} [11 - 3 + 2j - 1 - 3 - 2j] = \frac{1}{4} [4] = 1$$

which agrees with Fig. 4.12(b). The other values of $f(x)$ are obtained in a similar manner.

FIGURE 4.12
(a) A continuous function sampled ΔT units apart.
(b) Samples in the x -domain.
Variable t is continuous, while x is discrete.



DFT as Matrix Multiplication (computationally intensive)

$$F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M} \quad m = 0, 1, 2, \dots, M-1$$

$$\text{Let } \omega = e^{-j2\pi/M}$$

$$\begin{array}{c} \text{output} \\ \text{frequency} \\ \text{components} \end{array} \begin{bmatrix} F_0 \\ F_1 \\ F_2 \\ F_3 \\ \vdots \\ F_{M-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{M-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{2(M-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{3(M-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{M-1} & \omega^{2(M-1)} & \omega^{3(M-1)} & \dots & \omega^{(M-1)^2} \end{bmatrix} \begin{array}{c} \text{input} \\ \text{data} \end{array} \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \\ \vdots \\ f_{M-1} \end{bmatrix}$$

$[w^{ij}]$ ZERO-INDEXING

FAST FOURIER TRANSFORM

Assume the data vector f is of length 2^{k+1} .

1. Rearrange the input data vector f into two groups: even-indexed points and odd-indexed points.
2. The DFT matrix can be split into two smaller, simpler matrices:

$$\begin{bmatrix} F_1 \\ \vdots \\ F_{2^{k+1}} \end{bmatrix} = \begin{bmatrix} F_{2^{k+1}} \end{bmatrix} \begin{bmatrix} f_1 \\ \vdots \\ f_{2^{k+1}} \end{bmatrix} = \begin{bmatrix} I & D_w \\ I & D_w \end{bmatrix} \begin{bmatrix} F_{2^k} & 0 \\ 0 & F_{2^k} \end{bmatrix} \begin{bmatrix} f_{2^k}^{\text{even}} \\ f_{2^k}^{\text{odd}} \end{bmatrix}$$

where D_w = diagonal matrix of powers of w

I = identity matrix

3. The splitting process is applied recursively.

e.g. Let f be of size $1024 \Rightarrow 2^{k+1} = 1024 = 2^{10}$

$$F = F_{1024} \cdot f = \begin{bmatrix} I_{512} & D_{512} \\ I_{512} & D_{512} \end{bmatrix} \begin{bmatrix} F_{512} & 0 \\ 0 & F_{512} \end{bmatrix} \begin{bmatrix} f_{\text{even}} \\ f_{\text{odd}} \end{bmatrix} \quad \text{where } D_{512} = \begin{bmatrix} 1 & & & 0 \\ & w & & \\ & & w^2 & \\ & & & \ddots \\ 0 & & & & w^{511} \end{bmatrix}$$

$$F_{1024} \rightarrow F_{512} \rightarrow F_{256} \rightarrow \dots \rightarrow F_4 \rightarrow F_2$$

Even if your data is not of size $2^{k+1} \Rightarrow$ just pad zeroes

frequency index
 $m = 0, 1, 2, \dots, M-1$

$$\omega_m = \frac{2\pi m}{M}$$

$$F_m = \sum_{n=0}^{M-1} f_n \cdot e^{-j2\pi mn/M}$$

$$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m \cdot e^{j2\pi mn/M}$$

The DFT tells us how much of each frequency is present in the original signal.

F_m is COMPLEX.

- magnitude = $\sqrt{R^2 + I^2}$ = how much amplitude of that frequency is present
- angle = $\tan^{-1}(I/R)$ = the phase shift of that frequency component

The DFT takes M samples of the waveform f_n and produces an array F_m of length M .

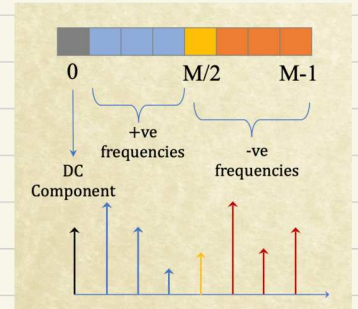
The array values are interpreted as follows:

- f_0 = corresponds to $M=0$ (no oscillation, just the constant offset in the signal)
- = the DC component
- = the average value of the signal

f_1 to $f_{M/2}$ = the positive frequency components

↳ corresponds to the Nyquist frequency

$f_{M/2+1}$ to f_{M-1} = the negative frequency components

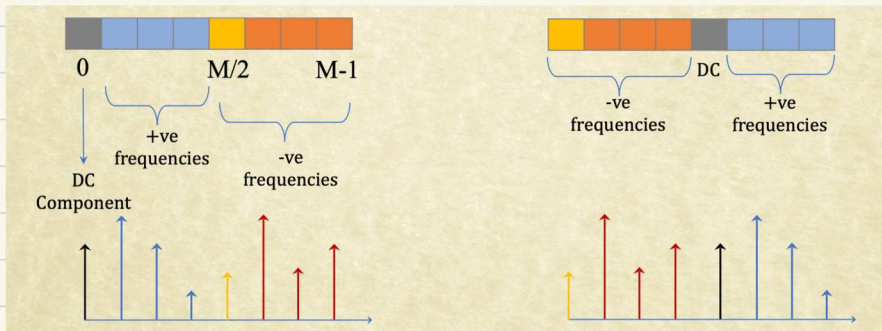


e.g. $M = 6$

$$f_0 = \text{DC}, \quad f_1 = \frac{f_s}{6}, \quad f_2 = \frac{2f_s}{6}, \quad f_3 = \frac{f_s}{2}, \quad f_4 = -\frac{2f_s}{6}, \quad f_5 = -\frac{f_s}{6}$$

The second half of the array represents phasors rotating in the opposite direction.

• For plotting, we first center-shift (rotate the array)



2-D IMPULSE AND ITS SIFTING PROPERTY

continuous: $\delta(t, z) = \begin{cases} 1 & \text{if } t = z = 0 \\ 0 & \text{otherwise} \end{cases}$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t, z) dt dz = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \cdot \delta(t, z) dt dz = f(0, 0)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \cdot \delta(t - t_0, z - z_0) dt dz = f(t_0, z_0)$$

discrete: $\delta(x, y) = \begin{cases} 1 & \text{if } x = y = 0 \\ 0 & \text{otherwise} \end{cases}$

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \cdot \delta(x, y) = f(0, 0)$$

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \cdot \delta(x - x_0, y - y_0) = f(x_0, y_0)$$

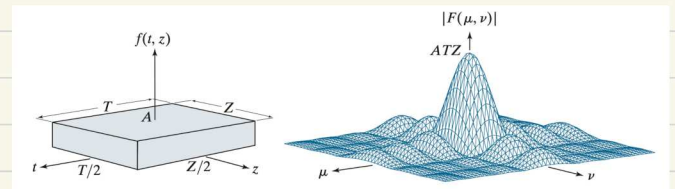
2D CONTINUOUS FOURIER TRANSFORM

$$F(\mu, \nu) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \cdot e^{-j2\pi(\mu t + \nu z)} dt dz$$

$$f(t, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mu, \nu) e^{j2\pi(\mu t + \nu z)} d\mu d\nu$$

e.g. Fourier transform of a 2D box function

$$\begin{aligned} F(\mu, \nu) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz = \int_{-T/2}^{T/2} \int_{-Z/2}^{Z/2} A e^{-j2\pi(\mu t + \nu z)} dt dz \\ &= ATZ \left[\frac{\sin(\pi \mu T)}{(\pi \mu T)} \right] \left[\frac{\sin(\pi \nu Z)}{(\pi \nu Z)} \right] \end{aligned}$$



2D DISCRETE FOURIER TRANSFORM

$$F_{m,n} = \sum_{y=0}^{N-1} \sum_{x=0}^{M-1} f_{x,y} \cdot e^{-j2\pi \left[\frac{mx}{M} + \frac{ny}{N} \right]}$$

$$f_{x,y} = \frac{1}{MN} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} F_{m,n} \cdot e^{j2\pi \left[\frac{mx}{M} + \frac{ny}{N} \right]}$$

slehar.com/www/fourier/fourier.html

EXAMPLES:

!!!

A single Fourier term encodes:

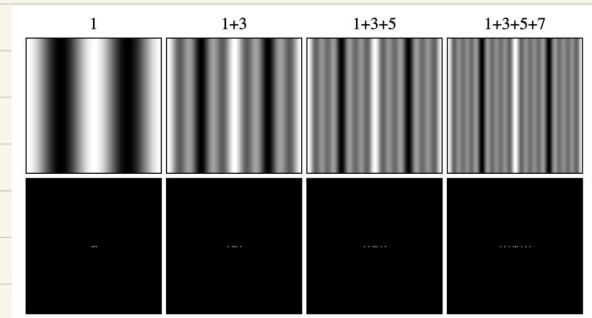
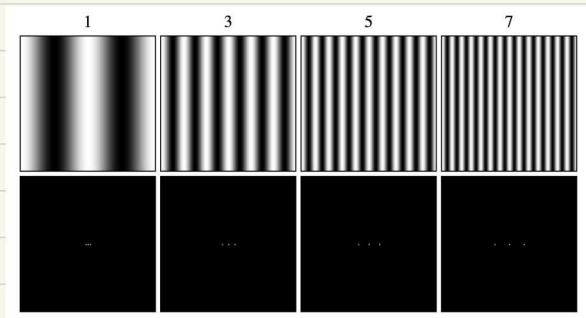
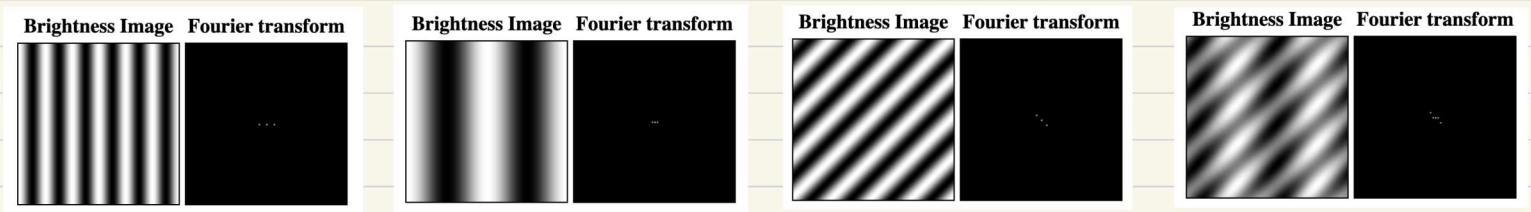
- spatial frequency = frequency across space with which the brightness modulates.
- magnitude = contrast = difference between the darkest and brightest peaks of the image.
- phase = how the wave is shifted relative to the origin.

Every pixel of the Fourier image is a spatial frequency value.

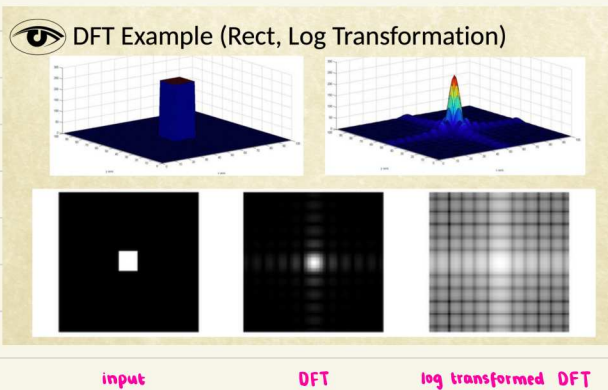
The magnitude of that value is encoded by the brightness of the pixel.

The pixel at the very center is the DC term.

average brightness across the whole image.

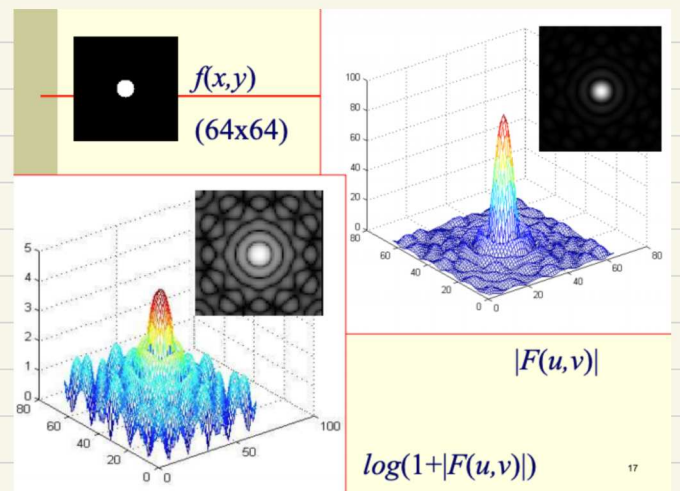
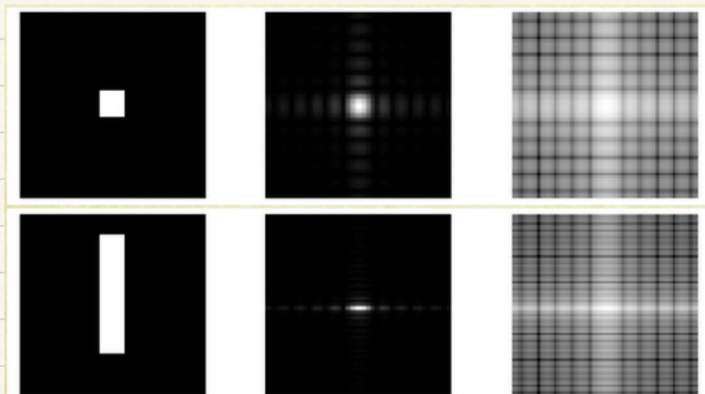


LOG TRANSFORMATION

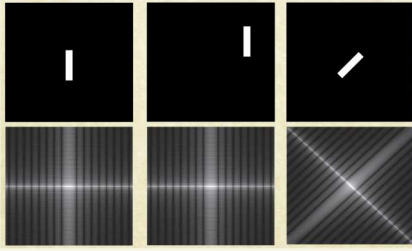


When viewing the Fourier transform, the DC component is often many orders of magnitude larger than the other frequency components.

A log transform compresses this large dynamic range, making the details of the higher frequency components much more visible.



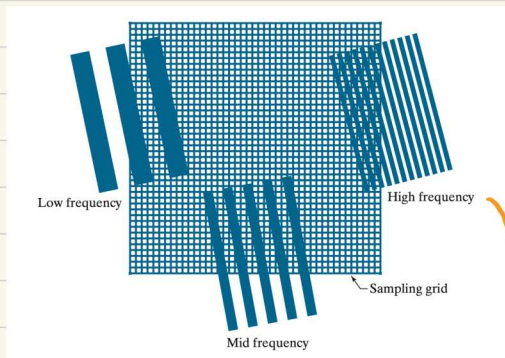
TRANSLATION / ROTATION



The spectrum is insensitive to image translation, but it rotates by the same angle of a rotated image.

SAMPLING FREQUENCY AND ALIASING

Superimposing the sampling grid on an image is analogous to multiplying the image by an impulse train. The sections of the regions inside the sampling grid are rough manifestations of how they would appear after sampling.



All 3 digitised regions exhibit aliasing to some extent.

Aliasing worsens as the discrepancy between detail (frequency) and sampling rate increasing.

Additional stripes appear and these stripes are rotated significantly w.r.t. the direction of the stripes in the continuous region. outside sampling grid

* Aliasing can also be a serious problem in image shrinking.

Shrinking an image $\equiv$ sampling the original image at a lower rate.



EXAMPLE 4.8: Illustration of aliasing in resampled natural images.

The effects of aliasing generally are worsened when the size of a digital image is reduced. Figure 4.19(a) is an image containing regions purposely selected to illustrate the effects of aliasing (note the thinly spaced parallel lines in all garments worn by the subject). There are no objectionable aliasing artifacts in Fig. 4.19(a), indicating that the sampling rate used initially was sufficient to mitigate visible aliasing.

In Fig. 4.19(b), the image was reduced to 33% of its original size using row/column deletion. The effects of aliasing are quite visible in this image (see, for example, the areas around scarf and the subject's knees). Images (a) and (b) are shown in the same size because the reduced image was brought back to its original size by pixel replication (the replication did not alter appreciably the effects of aliasing just discussed).

The digital "equivalent" of the defocusing of continuous images mentioned earlier for reducing aliasing, is to attenuate the high frequencies of a *digital* image by smoothing it with a lowpass filter before resampling. Figure 4.19(c) was processed in the same manner as Fig. 4.19(b), but the original image was smoothed using a 5×5 spatial averaging filter (see Section 3.5) before reducing its size. The improvement over Fig. 4.19(b) is evident. The image is slightly more blurred than (a) and (b), but aliasing is no longer objectionable.

MOIRÉ EFFECT

a b c
d e f

FIGURE 4.20

Examples of the moiré effect. These are vector drawings, not digitized patterns. Superimposing one pattern on the other is analogous to multiplying the patterns.

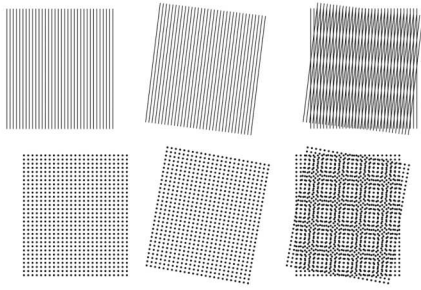
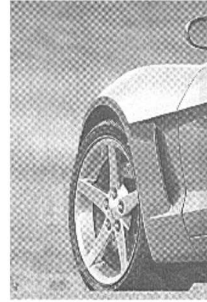


FIGURE 4.21

A newspaper image digitized at 75 dpi. Note the moiré-like pattern resulting from the interaction between the $\pm 45^\circ$ orientation of the half-tone dots and the north-south orientation of the sampling elements used to digitize the image.



A Moiré pattern is a secondary, visual phenomenon produced, for example, by superimposing two gratings of approximately equal spacing.

Separately, the patterns are clean and void of interference.

However, superimposing them creates a pattern with frequencies not present in either of the original patterns.

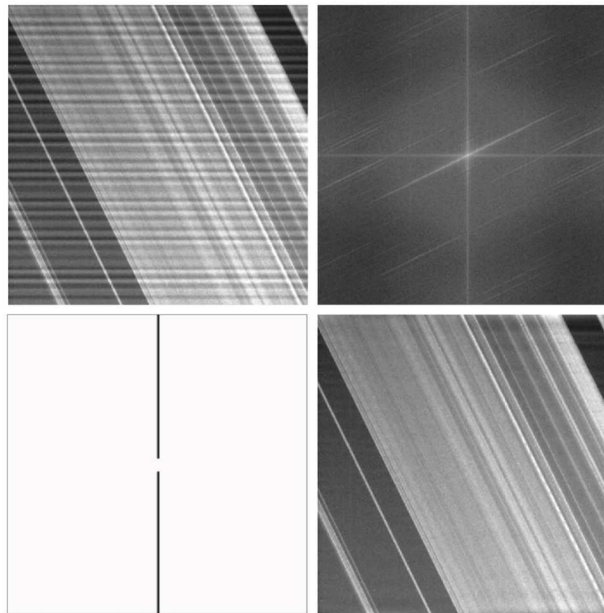
NOTCH FILTERS

= rejects or attenuates frequencies within a very narrow, specific range, while allowing all frequencies outside that range to pass through with minimal change.

a b
c d

FIGURE 4.65

(a) Image of Saturn rings showing nearly periodic interference. (b) Spectrum. (The bursts of energy in the vertical axis near the origin correspond to the interference pattern). (c) A vertical notch reject filter transfer function. (d) Result of filtering. (The thin black border in (c) is not part of the data.) (Original image courtesy of Dr. Robert A. West, NASA/JPL.)

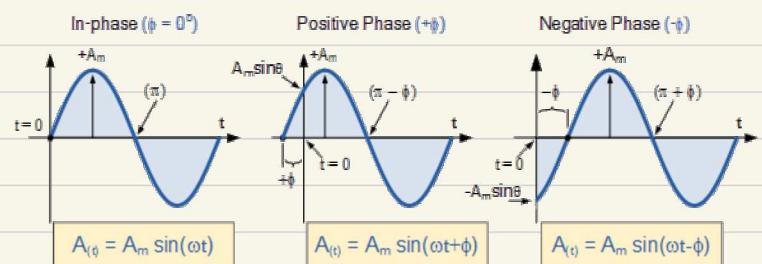


Important Terms

magnitude spectrum: $|F(u)| = \sqrt{R(u)^2 + I(u)^2}$

phase spectrum: $\Phi(u) = \tan^{-1}(I(u)/R(u))$

power spectrum: $P(u) = |F(u)|^2$



ingredients

recipe

MAGNITUDE AND PHASE SPECTRA



Figure 4a
Original

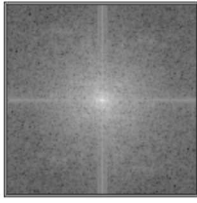


Figure 4b
 $\log(|A(\Omega, \Psi)|)$

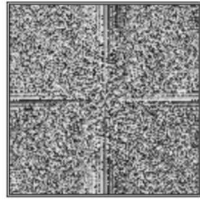


Figure 4c
 $\phi(\Omega, \Psi)$

magnitude spectrum
(log transformed)

phase spectrum

The phase spectrum contains the vast majority of the information needed to perceive the image's structure and details.

Phase information is far more critical for human visual perception than magnitude information.

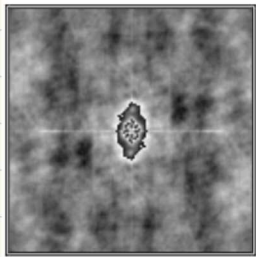


Figure 5a
 $\phi(\Omega, \Psi) = 0$



Figure 5b
 $|A(\Omega, \Psi)| = \text{constant}$

Fig. 5a

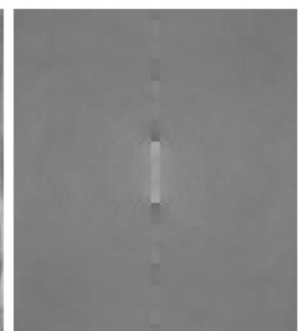
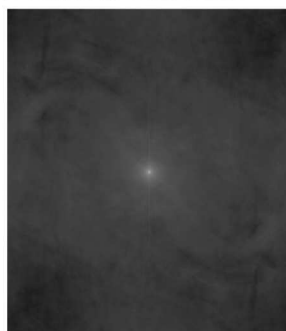
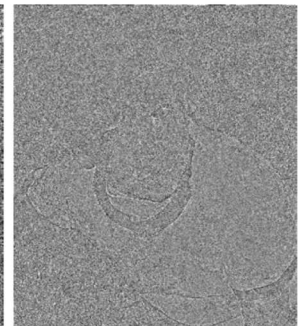
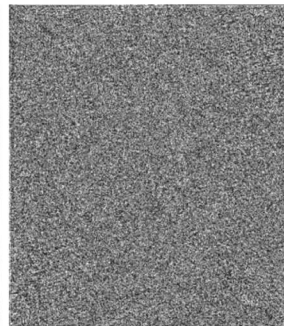
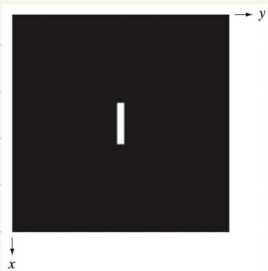
= IFT using magnitude spectrum but with phase spectrum = 0

Fig. 5b

= IFT using phase spectrum but with magnitude spectrum = constant

The phase spectrum is far more important than the magnitude spectrum for image recognition and perception.

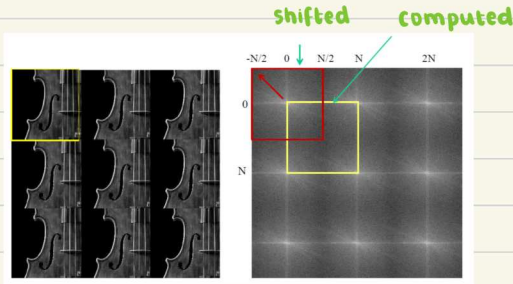
Fig 4.23(a)



a b c
d e f

FIGURE 4.26 (a) Boy image. (b) Phase angle. (c) Boy image reconstructed using only its phase angle (all shape features are there, but the intensity information is missing because the spectrum was not used in the reconstruction). (d) Boy image reconstructed using only its spectrum. (e) Boy image reconstructed using its phase angle and the spectrum of the rectangle in Fig. 4.23(a). (f) Rectangle image reconstructed using its phase and the spectrum of the boy's image.

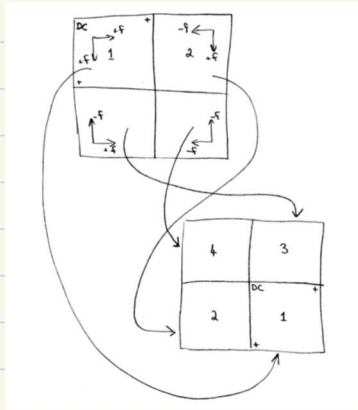
SHIFTING ORIGIN



STANDARD DFT

= the zero frequency (DC term) is at $(u,v) = (0,0)$ = top left corner

We need to shift the origin of the frequency spectrum to the center.



In 1D: 0, +ve $\rightarrow$ nyquist $\leftarrow$ -ve

In 2D: 0, +ve $\rightarrow$ nyquist $\leftarrow$ -ve

+ve
 $\downarrow$
nyquist
 $\uparrow$
-ve

IMAGE RECONSTRUCTION

- * Low frequencies represent the large-scale, smooth features of an image, like general shapes.
- * High frequencies represent the sharp details, textures and edges.

The low frequencies carry the most fundamental information about the image's content and structure.

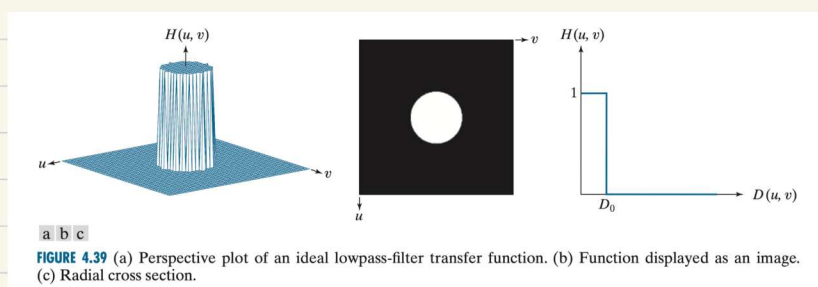
IDEAL LOWPASS FILTERS

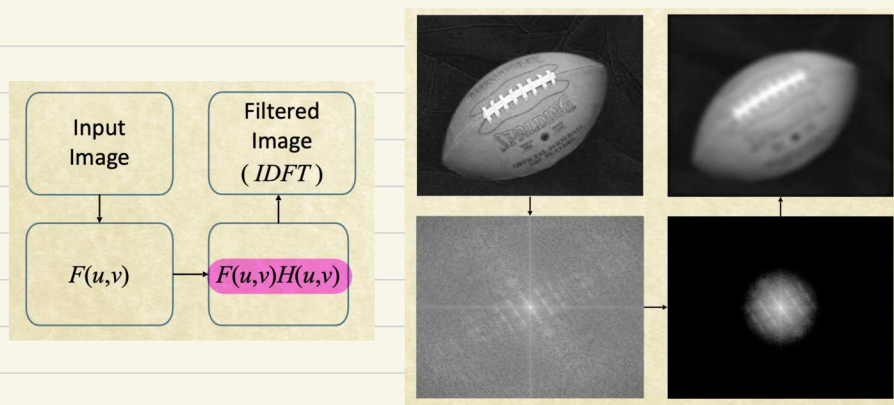
= passes without attenuation all frequencies within a circle of radius from the origin and "cuts off" all frequencies outside this

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

where D_0 = a +ve constant = cut-off frequency

$D(u,v)$ = distance b/w point (u,v) and the center of the $P \times Q$ frequency rectangle
 $= [(u-P/2)^2 + (v-Q/2)^2]^{1/2}$





One way to establish standard cutoff frequency loci is using circles that enclose specified amounts of total image power P_T , which we obtain by summing the components of the power spectrum at each point (u,v) .

$$P_T = \sum_{u=0}^{P-1} \sum_{v=0}^{Q-1} P(u,v)$$

If the DFT has been centered, a circle of radius D_0 with origin at the center of the frequency rectangle encloses α percent of the power.

$$\alpha = 100 \times \frac{\sum_u \sum_v P(u,v)}{P_T}$$

summation over values of (u,v) that lie inside the circle or on its boundary.

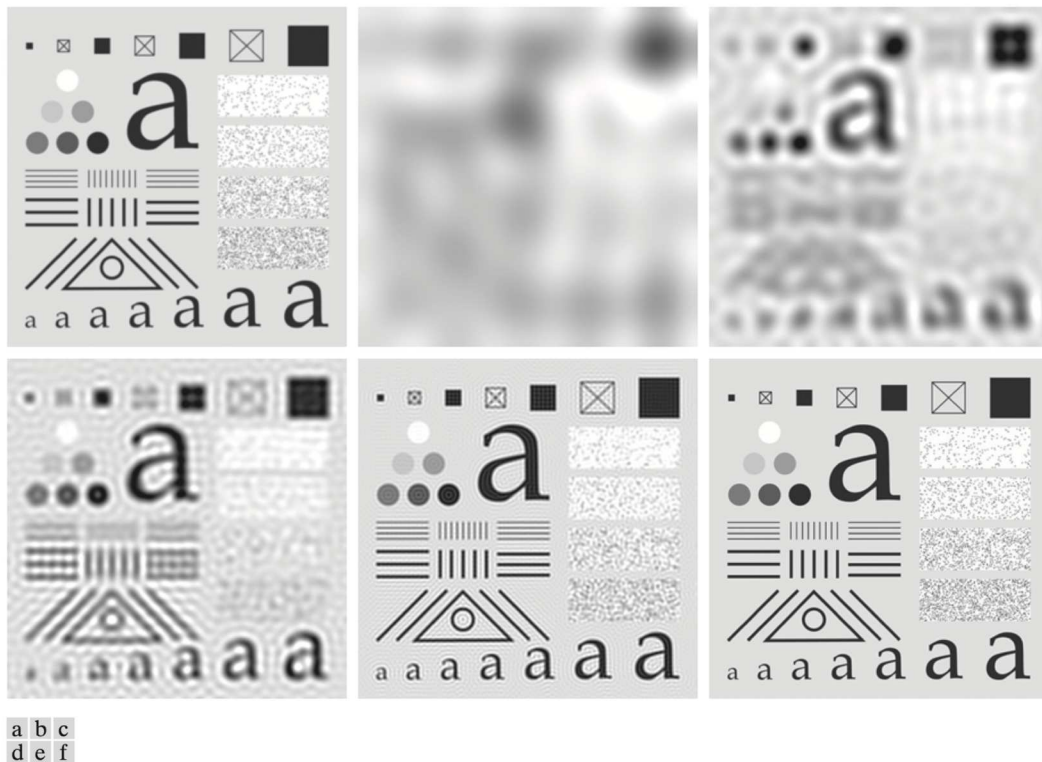
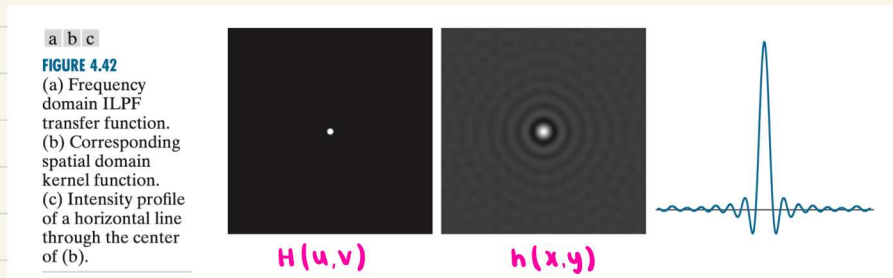


FIGURE 4.41 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.40(b). The power removed by these filters was 13.1, 7.2, 4.9, 2.4, and 0.6% of the total, respectively. We used mirror padding to avoid the black borders characteristic of zero padding, as illustrated in Fig. 4.31(c).

RINGING ARTIFACTS

The blurring and ringing properties of ILPFs can be explained using the convolution theorem.

$$f(x,y) * h(x,y) \Leftrightarrow F(u,v) H(u,v)$$

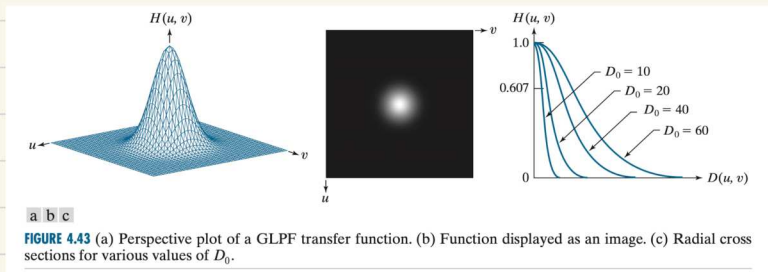


The center lobe of $h(x,y)$ is the principal cause of blurring, while the outer, smaller lobes are mainly responsible for ringing.

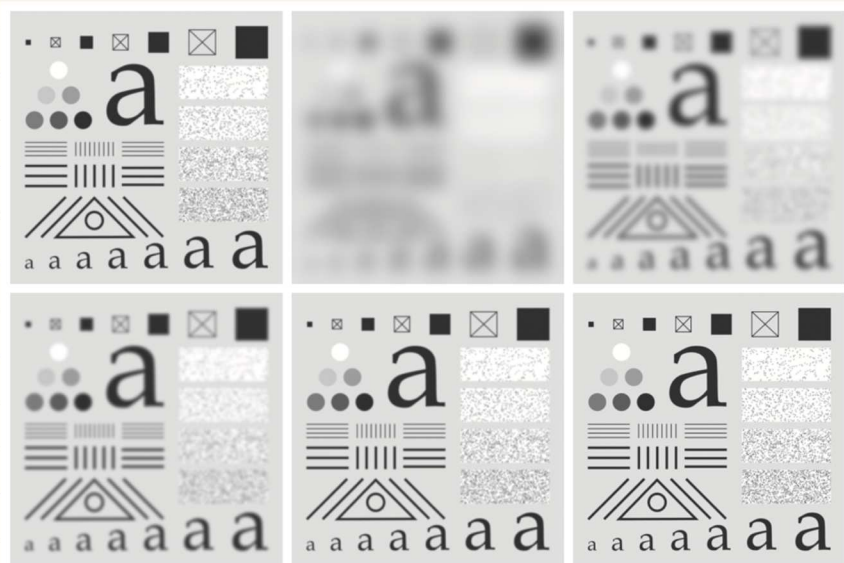
Because the "spread" of the spatial function is inversely proportional to the radius of $H(u,v)$,
 larger D_0 = more frequencies passed
 = the more the spatial function approaches an impulse

An impulse, in the limit, causes no blurring at all when convolved with the image.

GAUSSIAN LOWPASS FILTERS



$$H(u,v) = e^{\frac{-D(u,v)^2}{2D_0^2}}$$

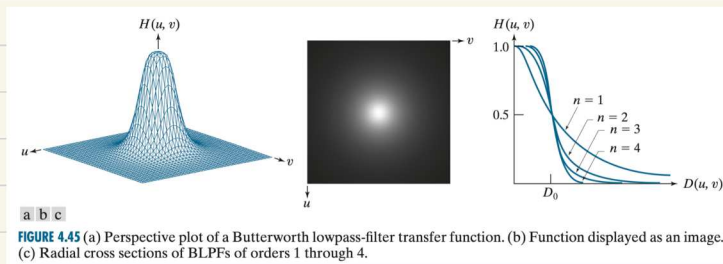


* Smoother transition in blurring as a function of increasing cutoff frequency.

* Slightly less smoothing

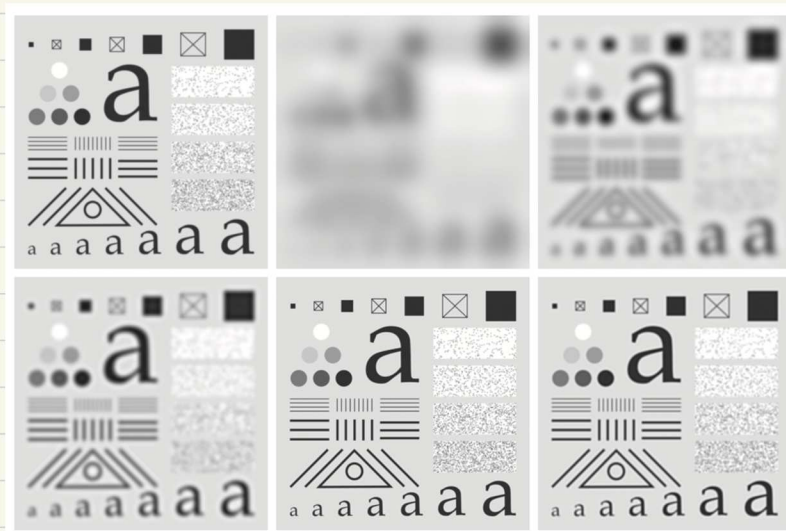
* KEY DIFFERENCE = NO RINGING

BUTTERWORTH LOWPASS FILTER



$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$$

n = order of the filter

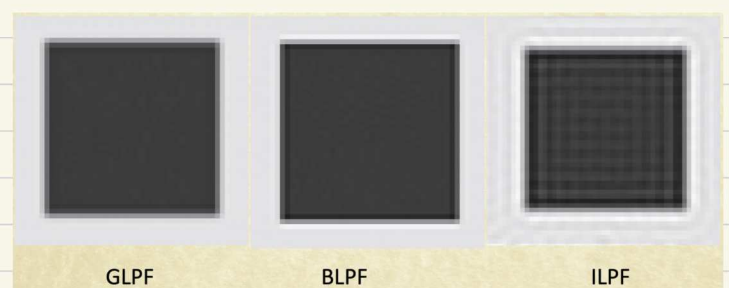
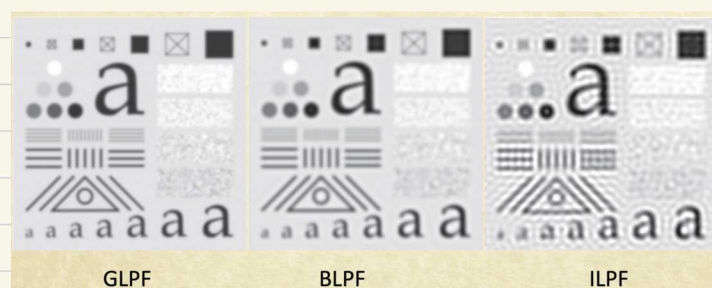


Ringing becomes significant for higher order filters.

Degree of blurring : ILPF > BLPF > GLPF

BLPF can give better control over cutoff with minimal ringing artifacts.

LPF COMPARISON



LPF APPLICATIONS

- ~ character recognition
- ~ increase resolution
- ~ reduce wrinkle lines
- ~ reduce scan lines

↗ for image sharpening

HIGHPASS FILTERING

$$H_{HP}(u,v) = 1 - H_{LP}(u,v)$$

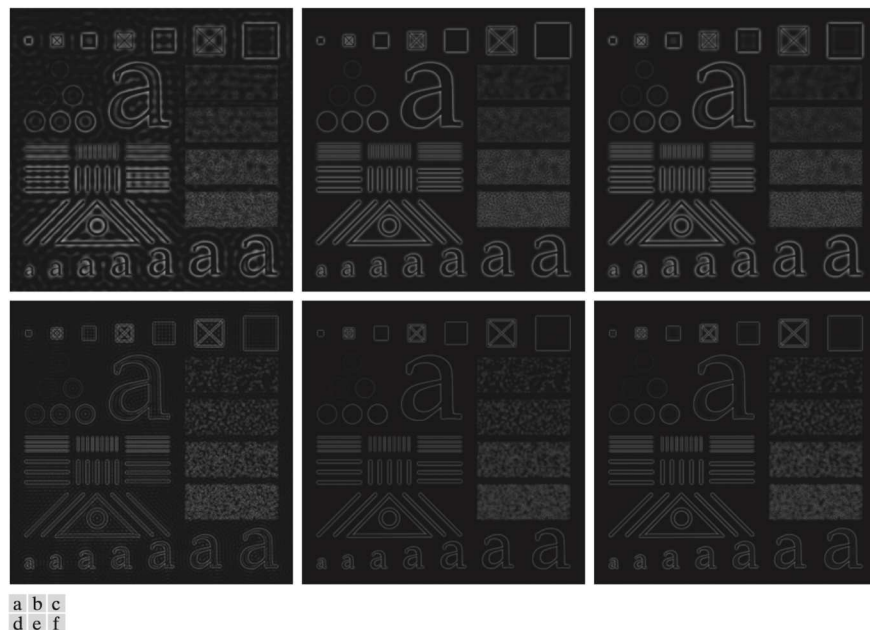
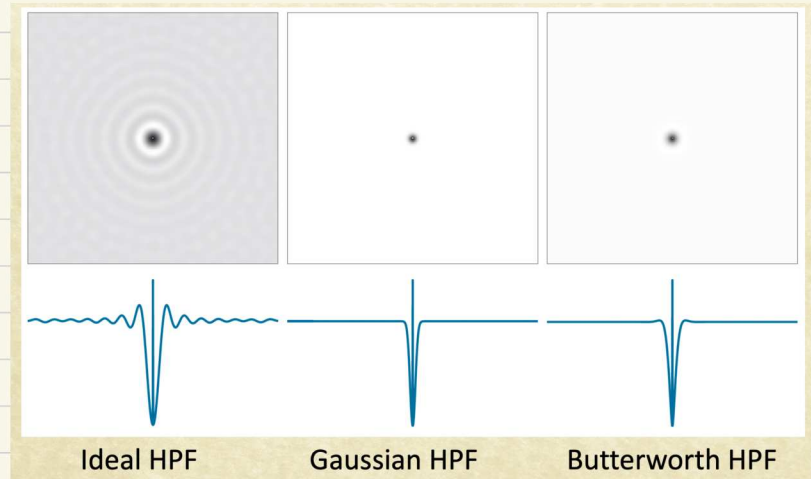
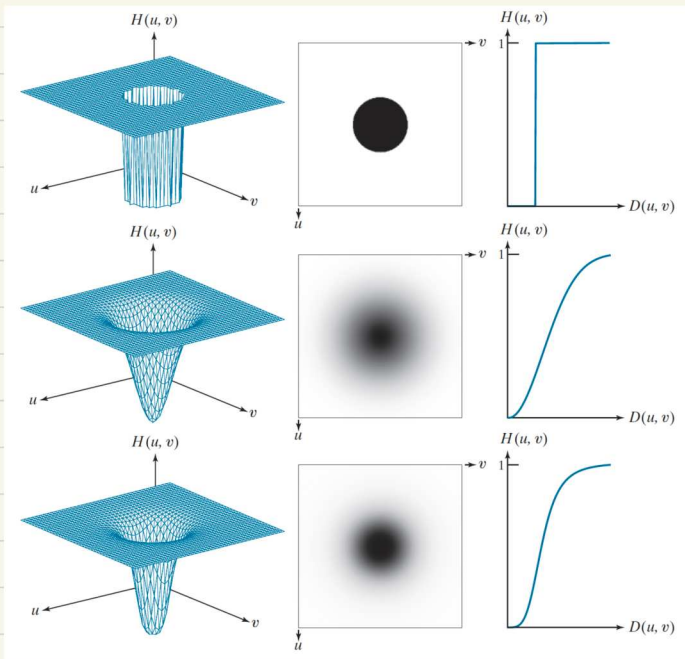


FIGURE 4.53 Top row: The image from Fig. 4.40(a) filtered with IHPF, GHPF, and BHPF transfer functions using $D_0 = 60$ in all cases ($n = 2$ for the BHPF). Second row: Same sequence, but using $D_0 = 160$.

APPLICATION : FINGERPRINT ENHANCEMENT

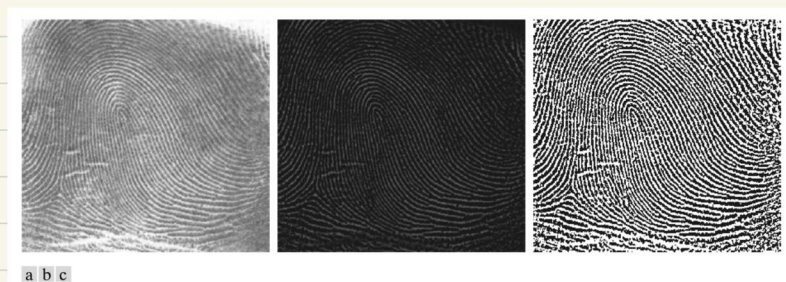


FIGURE 4.55 (a) Smudged thumbprint. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

LAPLACIAN IN THE FREQUENCY DOMAIN

$$F \left\{ \frac{\partial^n f(x)}{dx^n} \right\} = (ju)^n F(u)$$

$$F \left\{ \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2} \right\} = (ju)^2 F(u,v) + (jv)^2 F(u,v) = -(u^2 + v^2) F(u,v)$$

The Laplacian can be implemented in the frequency domain using the filter transfer function:

$$H(u,v) = -4\pi^2(u^2 + v^2)$$

Or, w.r.t to the center of the frequency rectangle:

$$\begin{aligned} H(u,v) &= -4\pi^2 \left[(u-P/2)^2 + (v-Q/2)^2 \right] \\ &= -4\pi^2 D^2(u,v) \end{aligned}$$

The Laplacian of an image $f(x,y)$ is obtained as:

$$\nabla^2 f(x,y) = F^{-1} \left\{ \underbrace{H(u,v) F(u,v)}_{\text{DFT of } f(x,y)} \right\}$$

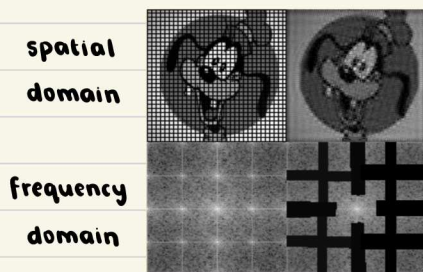
Enhancement is implemented using the equation:

$$g(x,y) = f(x,y) + c \cdot \nabla^2 f(x,y)$$

We take $c = -1$ since $H(u,v)$ is negative and we want a positive enhancement.

$$\begin{aligned} g(x,y) &= F^{-1} \{ F(u,v) - H(u,v) \cdot F(u,v) \} \\ &= F^{-1} \{ [1 - H(u,v)] F(u,v) \} \\ &= F^{-1} \{ [1 + 4\pi^2(u^2 + v^2)] F(u,v) \} \end{aligned}$$

ARTIFACT REMOVAL

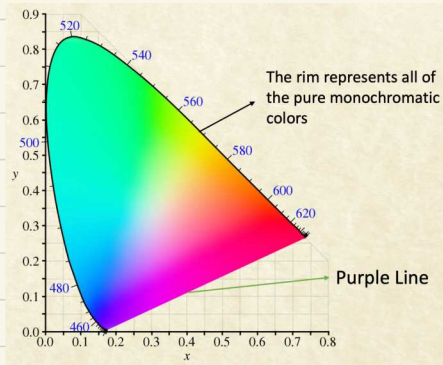


Apply a filter that removes the specific frequency components corresponding to the unwanted grid artifact.

CIE XYZ colour space

X, Y, Z = three imaginary primaries / axes
(based on human colour vision experiments)

We map to 2D to get the chromaticity diagram.



$$x = \frac{X}{X+Y+Z} \quad y = \frac{Y}{X+Y+Z} \quad z = \frac{Z}{X+Y+Z}$$

Since $x+y+z=1$
→ only two are independent

y (luminance) + xy (chromaticity)

* CIE Lab space is a perceptually uniform colour space derived from XYZ.

WHITE BALANCING

= to correct the overall colour cast in an image so that objects that should appear white actually look white, regardless of the colour of the light source

```
im = double(imread('lighthouse.jpg'));
```

```
RGBw = [246 169 87]; → measured colour of a reference white patch
```

```
im1(:,:,1) = im(:,:,1)*255/RGBw(1);  
im1(:,:,2) = im(:,:,2)*255/RGBw(2);  
im1(:,:,3) = im(:,:,3)*255/RGBw(3);
```

scaling pixel intensities for each channel
so that the white reference becomes [255, 255, 255]

Von Kries Method

= a biologically motivated model

= colour constancy is achieved by individually adapting the gains of the three cone responses

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 1/L'_w & 0 & 0 \\ 0 & 1/M'_w & 0 \\ 0 & 0 & 1/S'_w \end{bmatrix} \begin{bmatrix} L' \\ M' \\ S' \end{bmatrix}$$

L', M', S' = cone responses under original illumination

L'_w, M'_w, S'_w = cone responses of the reference white under that illumination

L, M, S = adapted cone responses after white balancing

$$L = L' / L'_w \quad M = M' / M'_w \quad S = S' / S'_w$$

Scale each channel so that the white point becomes neutral.

IMAGE RESTORATION

$$g(x,y) = \mathcal{H}\{f(x,y)\} + \eta(x,y)$$

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

$$G(u,v) = H(u,v) F(u,v) + N(u,v)$$

Restoration in the presence of noise only

ADAPTIVE LOCAL NOISE REDUCTION FILTER

$$\hat{f}(x,y) = g(x,y) - \frac{\sigma_n^2}{\sigma_L^2} [g(x,y) - m_L]$$

σ_n = variance of the noise

σ_L = local variance (high around edges)

m_L = local mean

$$g(x,y) > m_L \Rightarrow \frac{\sigma_n^2}{\sigma_L^2} [g(x,y) - m_L] \text{ will be subtracted from } g(x,y)$$
$$\Rightarrow g(x,y) \text{ will be pulled down closer to the mean}$$

$$g(x,y) < m_L \Rightarrow \frac{\sigma_n^2}{\sigma_L^2} [g(x,y) - m_L] \text{ will be added to } g(x,y)$$
$$\Rightarrow g(x,y) \text{ will be pulled up closer to the mean}$$

$$\sigma_L \gg \sigma_n \Rightarrow \text{presence of an edge} \Rightarrow \text{less pull}$$

Notch Filters

• NOTCH REJECT

$$H_{NR} = \prod_{k=1}^Q H_k(u,v) \cdot H_{-k}(u,v)$$

removes noise spike at (u_k, v_k)

removes noise spike at $(-u_k, -v_k)$

Periodic noise in a real image appears as a pair of conjugate spikes in the frequency domain.

Q = no. of periodic noise components (spikes) to remove.

Multiplying the individual filters creates a composite filter that rejects all targeted frequencies simultaneously.

• NOTCH PASS

$$H_{NP} = 1 - H_{NR}$$

OPTIMUM NOTCH FILTERING

$$\eta(x,y) = \mathcal{F}^{-1} \{ H_{NP}(u,v) \cdot G(u,v) \}$$

↳ estimated periodic noise image.

$$\hat{f}(x,y) = g(x,y) - w(x,y) \cdot \eta(x,y)$$

↳ weighting function

$$w = \frac{\overline{g\eta} - \bar{g}\bar{\eta}}{\overline{\eta^2} - \bar{\eta}^2}$$

COVARIANCE

VARIANCE

How much does g look like η , compared to how strong η is?

g and η strongly correlated $\Rightarrow$ large $w \Rightarrow$ subtract more of η

g and η weakly correlated $\Rightarrow$ small $w \Rightarrow$ subtract less of η

WEINER FILTER

We want a function $W(u,v)$ such that $\hat{F}(u,v) = W(u,v) G(u,v)$

and it minimises $MSE = E\{(f-\hat{f})^2\} / E\{(f-\hat{f})^2\}$

We know $G = HF + N$

$$E = F - \hat{F} = F - WG = F - W(HF + N) = (I - WH)F - WN$$

$$\therefore MSE = E\{E^2\} = E\{|(I - WH)F - WN|^2\}$$

Assumption: F and N are not correlated.

$S_F(u,v) = E\{|F|^2\}$ = power spectral density of image

$S_N(u,v) = E\{|N|^2\}$ = power spectral density of noise

$$\therefore MSE = |I - WH|^2 S_F + |W|^2 S_N$$

We want to minimise MSE wrt W

$$\frac{\partial MSE}{\partial W} = -H(I - WH)S_F + WS_N = 0$$

$$H^*(I - WH)S_F = WS_N \Rightarrow H^*S_F = W(HH^*S_F + S_N) \Rightarrow W = \frac{H^*S_F}{|H|^2S_F + S_N} \therefore W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + S_N/S_F}$$

↙
can replace with K

* No noise $\Rightarrow S_N = 0 \Rightarrow W = \frac{H^*}{|H|^2} = \frac{1}{H} \Rightarrow \hat{F} = \frac{G}{H}$ (inverse filter)

* Noise dominates $\Rightarrow S_N \gg S_F \Rightarrow W = \frac{H^* S_F}{S_N} \approx 0$

($G \approx N \Rightarrow$ best guess w.r.t MSE is $\hat{F} = 0$)

$\therefore$ Wiener filter can handle both degradation and noise.

NOTES

1. COLOUR QUANTISATION

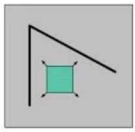
if you divide the entire colour cube into uniform cubelets (e.g. $4 \times 4 \times 4$), most of them will be empty
 $\therefore$ use k-means
 $\Rightarrow$ quantise the space actually occupied by the pixels.

2. COLOUR EDGE DETECTION

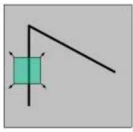
if you do edge detection in HSI space, colour change might not be detected if the intensities of the colour are the same.

HARRIS CORNER DETECTION

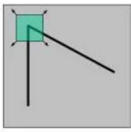
corner $\Rightarrow$ intensity changes sharply in multiple directions



"flat" region:
no change in all
directions



"edge":
no change along the
edge direction



"corner":
significant change in
all directions

* a flat region has almost no intensity change

* an edge has intensity change in only one direction.

* a corner has significant intensity change in both x and y directions.

Imagine taking a small window (patch) around a pixel and shifting it slightly in various directions.

A corner is where the local intensity pattern changes the most in all directions.

Let's define intensity at pixel (x, y) as $I(x, y)$.

We consider the change in intensity when the window shifts by (u, v) .

$$E(u, v) = \sum_{x, y} \underbrace{w(x, y)}_{\substack{\text{window function to weigh nearby pixels more}}} [I(x+u, y+v) - I(x, y)]^2$$

For small shifts, we can approximate with Taylor expansion:

$$I(x+u, y+v) \approx I(x, y) + I_x u + I_y v$$

$$\Rightarrow E(u, v) \approx \sum_{x, y} w(x, y) [I_x u + I_y v]^2 \approx \sum_{x, y} w(x, y) [I_x^2 u^2 + 2 I_x I_y uv + I_y^2 v^2]$$

$$\Rightarrow E(u, v) \approx \begin{bmatrix} u & v \end{bmatrix} M \begin{bmatrix} u \\ v \end{bmatrix} \quad \text{where } M = \sum_{x, y} w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

* The eigenvectors of M give the directions of principal change.

* The eigenvalues tell you how fast the change happens in those directions.

Computing eigenvalues for every pixel is expensive.

Harris approximates the result using:

$$R = \det(M) - k \cdot (\text{trace}(M))^2$$

$\left\{ \begin{array}{l} \text{large +ve} \Rightarrow \text{corner} \\ \text{near zero} \Rightarrow \text{edge} \\ \text{small / negative} \Rightarrow \text{flat region} \end{array} \right.$

$$\det(M) = \lambda_1 \lambda_2$$

$$\text{trace}(M) = \lambda_1 + \lambda_2$$

large determinant $\Rightarrow$ both eigenvalues large $\Rightarrow$ corner

large trace, small determinant $\Rightarrow$ one eigenvalue large, one small $\Rightarrow$ edge

both small $\Rightarrow$ flat